

Accelerator ML Living Review

Summary Statistics

per_year: 12
per_category: 16
per_venue/journal: 234
per_keyword: 17
monthly_trends: 82

Papers

Machine learning as a service system for particle accelerator and its application in CSNS

Mei, Hao, Zhang, Yuliang, Peng, Na, Cheng, Sinong, He, Yongcheng, Xue, Kangjia, Wang, Lin, Li, Mingtao, Wu, Xuan, Zhu, Peng (2025)

Radiation Detection Technology and Methods

A generative adversarial network to improve integrated mode proton imaging resolution using paired proton-carbon data.

Simard M, Fullarton R, Volz L, Schuy C, Chung S, Baker C, Graeff C, Fekete CC. (2025)

Med Phys

A Time-Series Approach for Machine Learning-Based Patient-Specific Quality Assurance of Radiosurgery Plans

Buzzi S, Mancosu P, Bresolin A, Gallo P, La Fauci F, Lobefalo F, Paganini L, Pelizzoli M, Reggiori G, Franzese C, Tomatis S, Scorsetti M, Lenardi C, Lambri N. (2025)

Bioengineering (Basel)

A Supervised Machine Learning Framework for Multipactor Breakdown Prediction in High-Power Radio Frequency Devices and Accelerator Components: A Case Study in Planar Geometry

Asif Iqbal, John Verboncoeur, Peng Zhang (2025)

arXiv

Geoff: The Generic Optimization Framework & Frontend for Particle Accelerator Controls

Penelope Madysa, Sabrina Appel, Verena Kain, Michael Schenk (2025)

arXiv

Towards Agentic AI on Particle Accelerators

Antonin Sulc, Thorsten Hellert, Raimund Kammering, Hayden Hoschouer, Jason St. John (2025)

arXiv

Acceleration of Multi-Scale LTS Magnet Simulations with Neural Network Surrogate Models

Louis Denis, Julien Dular, Vincent Nuttens, Mariusz Wozniak, Benoît Vanderheyden, Christophe Geuzaine (2025)

arXiv

Application Of Large Language Models For The Extraction Of Information From Particle Accelerator Technical Documentation

Qing Dai, Rasmus Ischebeck, Maruisz Sapinski, Adam Grycner (2025)

arXiv

Predictive Hydrodynamic Simulations for Laser Direct-drive Implosion Experiments via Artificial Intelligence

Zixu Wang, Yuhan Wang, Junfei Ma, Fuyuan Wu, Junchi Yan, Xiaohui Yuan, Zhe Zhang, Jie Zhang (2025)

arXiv

Towards Foundation Models for Experimental Readout Systems Combining Discrete and Continuous Data

James Giroux, Cristiano Fanelli (2025)

arXiv

Towards generalizable deep ptychography neural networks

Albert Vong, Steven Henke, Oliver Hoidn, Hanna Ruth, Junjing Deng, Alexander Hexemer, Apurva Mehta, Arianna Gleason, Levi Hancock, Nicholas Schwarz (2025)

arXiv

FusionMAE: large-scale pretrained model to optimize and simplify diagnostic and control of fusion plasma

Zongyu Yang, Zhenghao Yang, Wenjing Tian, Jiyuan Li, Xiang Sun, Guohui Zheng, Songfen Liu, Niannian Wu, Rongpeng Li, Zhaohe Xu, Bo Li, Zhongbing Shi, Zhe Gao, Wei Chen, Xiaoquan Ji, Min Xu, Wulyu Zhong (2025)

arXiv

A Surrogate model for High Temperature Superconducting Magnets to Predict Current Distribution with Neural Network

Mianjun Xiao, Peng Song, Yulong Liu, Cedric Korte, Ziyang Xu, Jiale Gao, Jiaqi Lu, Haoyang Nie, Qiantong Deng, Timing Qu (2025)

arXiv

Machine Learning approach to classifying quench antenna signals

Plebani, Alberto, Barzi, Emanuela, Teyber, Reed (2025)

Thesis

Machine Learning for superconducting magnets application

Stabilini, Elisa (2025)

Thesis

Developing Machine Learning Models for Proton Computed Tomography and LHCb Particle Tracking

Ackernley, Thomas (2025)

Thesis

Batch spacing optimization by reinforcement learning

Remta, Matthias, Velotti, Francesco, Rezagholi, Sharwin (2025)

Phys.Rev.Accel.Beams

Road map for the tuning of hadronic interaction models with accelerator-based and astroparticle data

Albrecht, J., Becker Tjus, J., Behling, N., Blazek, J., Bleicher, M., Boelhauve, J., Cazon, L., Conceicao, R., Dembinski, H., Dietrich, L., Ebr, J., Ellbracht, J., Engel, R., Fedynitch, A., Fieg, M., Garzelli, M.V., Gaudu, C., Graziani, G., Gutjahr, P., Haungs, A., Huege, T., Hymon, K., Hünnefeld, Mirco, Kampert, K.-H., Kardum, L., Kolk, L., Korneeva, N., Kröninger, K., Maire, A., Menjo, H., Morejon, L., Ostapchenko, S., Paakkinen, P., Pierog, T., Plotko, P., Prosekin, A., Pyras, L., Pöschl, T., Rautenberg, Julian, Reininghaus, M., Rhode, W., Riehn, F., Roth, M., Sandrock, A., Sarcevic, I., Schmelling, M., Sigl, G., Sjöstrand, T., Soldin, D., Unger, M., Utheim, M., Vícha, J., Werner, K., Windau, M.E., Zhukov, V. (2025)

arXiv

Self-supervised physics-informed generative networks for phase retrieval from a single X-ray hologram

Yang, Xiaogang, Hailu, Dawit, Kulvait, Vojtech, Jentschke, Thomas, Flenner, Silja, Greving, Imke, Campbell, Stuart I., Hagemann, Johannes, Schroer, Christian G., Wong, Tak Ming, Moosmann, Julian (2025)

Opt.Express

A Supervised Machine Learning Framework for Multipactor Breakdown Prediction in High-Power Radio Frequency Devices and Accelerator Components: A Case Study in Planar Geometry

Iqbal, Asif, Verboncoeur, John, Zhang, Peng (2025)
arXiv

Analysis note: measurement of thrust in e^+e^- collisions at $\sqrt{s} = 91$ GeV with archived ALEPH data

Badea, Anthony, Baty, Austin, Bossi, Hannah, Chen, Yu-Chen, Chen, Yi, Zhang, Jingyu, Innocenti, Gian Michele, Maggi, Marcello, McGinn, Chris, Peters, Michael, Sheng, Tzu-An, Mikuni, Vinicius, Avaylon, Matthew, Komiske, Patrick, Metodiev, Eric, Thaler, Jesse, Nachman, Benjamin, Lee, Yen-Jie (2025)
arXiv

Machine learning application for particle accelerator optimization-a review

Isti Dian Rachmawati, Nazrul Effendy, Taufik Taufik (2025)
IAES International Journal of Artificial Intelligence

Machine Learning for Particle Accelerators

Elena Fol, Auralee Edelen (2025)
Unknown Venue

FORT-GCN: A Fault-tolerant and Adaptive Accelerator Design for Efficient Graph Convolutional Network Inference

Ke Wang, Yingnan Zhao, Ahmed Louri (2025)
ACM Transactions on Embedded Computing Systems

(Invited) Analog Signal Processing Technologies for Power-Efficient Neural Network Inference and Training

Bert Jan Offrein (2025)
Meeting abstracts/Meeting abstracts (Electrochemical Society. CD-ROM)

Machine learning based parametrization of the resolution function for the first experimental area of the n_TOF facility at CERN

Petar Žugec, Marta Sabaté-Gilarte, Michael Bacak, Vasilis Vlachoudis, Adria Casanovas, Francisco García-Infantes (2025)
Nuclear Science and Techniques

-Diagnosis of Nasopalatine Duct and Nasopalatine Duct Cyst in CBCT Images: A Radiomics-Based Machine Learning Approach

H Duyan Yüksel, B Büyük, B Evlice (2025)
Dentomaxillofacial Radiology

Integration of Machine Learning-Based Plasma Acceleration Simulations into Geant4: A Case Study with the PALLAS Experiment

A. Sytov, K. Cassou, V. Kubyskyi, M. Lenivenko, A. Huber (2025)
arXiv

Optimisation of the Accelerator Control by Reinforcement Learning: A Simulation-Based Approach

Anwar Ibrahim, Denis Derkach, Alexey Petrenko, Fedor Ratnikov, Maxim Kaledin (2025)
arXiv

Physics-Informed Super-Resolution Diffusion for 6D Phase Space Diagnostics

Alexander Scheinker (2025)
arXiv

Optimizing Beam-Plasma Interactions Through Jitter Analysis Using Start-to-End Simulations

Robin Hwang (2024)
arXiv

Virtual Pulse Reconstruction Diagnostic for Single-Shot Measurement of Free Electron Laser Radiation Power

Till Korten, Vladimir Rybnikov, Peter Steinbach, Najmeh Mirian (2024)

arXiv

Harnessing Machine Learning for Single-Shot Measurement of Free Electron Laser Pulse Power

Till Korten, Vladimir Rybnikov, Mathias Vogt, Juliane Roensch-Schulenburg, Peter Steinbach, Najmeh Mirian (2024)

arXiv

Data-Driven Gradient Optimization for Field Emission Management in a Superconducting Radio-Frequency Linac

Steven Goldenberg, Kawser Ahammed, Adam Carpenter, Jiang Li, Riad Suleiman, Chris Tennant (2024)

arXiv

Data-Driven Discovery of Beam Centroid Dynamics

Liam A. Pocher, Irving Haber, Thomas M. Antonsen Jr., Patrick G. O'Shea (2024)

arXiv

Design and development of advanced Al-Ti-V alloys for beampipe applications in particle accelerators

Kamaljeet Singh, Kangkan Goswami, Raghunath Sahoo, Sumanta Samal (2025)

arXiv

Machine Learning for Reducing Noise in RF Control Signals at Industrial Accelerators

M. Henderson, J. P. Edelen, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2024)

arXiv

Surrogate Models studies for laser-plasma accelerator electron source design through numerical optimisation

G. Kane, P. Drobniak, S. Kazamias, V. Kubytskyi, M. Lenivenko, B. Lucas, J. Serhal, K. Cassou, A. Beck, A. Specka, F. Massimo (2025)

arXiv

Beamline Steering Using Deep Learning Models

Dexter Allen, Isaac Kante, Dorian Bohler (2024)

arXiv

Beam-based Identification of Magnetic Field Errors in a Synchrotron using Deep Lie Map Networks

Conrad Caliri, Adrian Oeftiger, Oliver Boine-Frankenheim (2024)

arXiv

Linac_Gen: integrating machine learning and particle-in-cell methods for enhanced beam dynamics at Fermilab

Abhishek Pathak (2024)

arXiv

Automated Anomaly Detection on European XFEL Klystrons

Antonin Sulc, Annika Eichler, Tim Wilksen (2024)

arXiv

Accelerator beam phase space tomography using machine learning to account for variations in beamline components

Andrzej Wolski, Diego Botelho, David Dunning, Amelia E. Pollard (2024)

arXiv

Large Language Models for Human-Machine Collaborative Particle Accelerator Tuning through Natural Language

Jan Kaiser, Annika Eichler, Anne Lauscher (2024)

arXiv

Accelerating Cavity Fault Prediction Using Deep Learning at Jefferson Laboratory

Monibor Rahman, Adam Carpenter, Khan Iftekharuddin, Chris Tennant (2024)

arXiv

Anomaly Detection of Particle Orbit in Accelerator using LSTM Deep Learning Technology

Zhiyuan Chen, Wei Lu, Radhika Bhong, Yimin Hu, Brian Freeman, Adam Carpenter (2024)

arXiv

Machine-learning approach for operating electron beam at KEK S_e^-/e^+ injector Linac

Gaku Mitsuka, Shinnosuke Kato, Naoko Iida, Takuya Natsui, Masanori Satoh (2024)

arXiv

Cheetah: Bridging the Gap Between Machine Learning and Particle Accelerator Physics with High-Speed, Differentiable Simulations

Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia (2024)

arXiv

Beyond PID Controllers: PPO with Neuralized PID Policy for Proton Beam Intensity Control in Mu2e

Chenwei Xu, Jerry Yao-Chieh Hu, Aakaash Narayanan, Mattson Thieme, Vladimir Nagaslaev, Mark Austin, Jeremy Arnold, Jose Berlioz, Pierrick Hanlet, Aisha Ibrahim, Dennis Nicklaus, Jovan Mitrevski, Jason Michael St. John, Gauri Pradhan, Andrea Saewert, Kiyomi Seiya, Brian Schupbach, Randy Thurman-Keup, Nhan Tran, Rui Shi, Seda Ogrenci, Alexis Maya-Isabelle Shuping, Kyle Hazelwood, Han Liu (2023)

arXiv

Robust Errant Beam Prognostics with Conditional Modeling for Particle Accelerators

Kishansingh Rajput, Malachi Schram, Willem Blokland, Yasir Alanazi, Pradeep Ramuhalli, Alexander Zhukov, Charles Peters, Ricardo Vilalta (2024)

arXiv

Machine Learning For Beamline Steering

Isaac Kante (2023)

arXiv

Variational Autoencoders for Noise Reduction in Industrial LLRF Systems

J. P. Edelen, M. J. Henderson, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2023)

arXiv

Resilient VAE: Unsupervised Anomaly Detection at the SLAC Linac Coherent Light Source

Ryan Humble, William Collocho, Finn O'Shea, Daniel Ratner, Eric Darve (2023)

arXiv

Time-drift Aware RF Optimization with Machine Learning Techniques

R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2023)

arXiv

Distance Preserving Machine Learning for Uncertainty Aware Accelerator Capacitance Predictions

Steven Goldenberg, Malachi Schram, Kishansingh Rajput, Thomas Britton, Chris Pappas, Dan Lu, Jared Walden, Majdi I. Radaideh, Sarah Cousineau, Sudarshan Harave (2023)

arXiv

Learning to Do or Learning While Doing: Reinforcement Learning and Bayesian Optimisation for Online Continuous Tuning

Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia, Oliver Stein, Erik Bründermann, Willi Kuroпка, Hannes Dinter, Frank Mayet, Thomas Vinatier, Florian Burkart, Holger Schlarb (2023)

arXiv

From Compact Plasma Particle Sources to Advanced Accelerators with Modeling at Exascale

Axel Huebl, Remi Lehe, Edoardo Zoni, Olga Shapoval, Ryan T. Sandberg, Marco Garten, Arianna

Formenti, Revathi Jambunathan, Prabhat Kumar, Kevin Gott, Andrew Myers, Weiqun Zhang, Ann Almgren, Chad E. Mitchell, Ji Qiang, David Grote, Alexander Sinn, Severin Diederichs, Maxence Thevenet, Luca Fedeli, Thomas Clark, Neil Zaim, Henri Vincenti, Jean-Luc Vay (2023)
arXiv

Forecasting Particle Accelerator Interruptions Using Logistic LASSO Regression

Sichen Li, Jochem Snuerink, Fernando Perez-Cruz, Andreas Adelman (2023)

arXiv

Learning Electron Bunch Distribution along a FEL Beamline by Normalising Flows

Anna Willmann, Jurjen Couperus Cabada, Yen-Yu Chang, Richard Pausch, Amin Ghaith, Alexander Debus, Arie Irman, Michael Bussmann, Ulrich Schramm, Nico Hoffmann (2023)

arXiv

Physics-constrained 3D Convolutional Neural Networks for Electrodynamics

Alexander Scheinker, Reju Pokharel (2023)

arXiv

Identification of Magnetic Field Errors in Synchrotrons based on Deep Lie Map Networks

Conrad Caliri, Adrian Oeftiger, Oliver Boine-Frankenheim (2023)

arXiv

Data-driven Science and Machine Learning Methods in Laser-Plasma Physics

Andreas Döpp, Christoph Eberle, Sunny Howard, Faran Irshad, Jinpu Lin, Matthew Streeter (2023)

arXiv

Applications of Differentiable Physics Simulations in Particle Accelerator Modeling

Ryan Roussel, Auralee Edelen (2022)

arXiv

Prior-mean-assisted Bayesian optimization application on FRIB Front-End tuning

Kilean Hwang, Tomofumi Maruta, Alexander Plastun, Kei Fukushima, Tong Zhang, Qiang Zhao, Peter Ostroumov, Yue Hao (2022)

arXiv

Fault Prognosis in Particle Accelerator Power Electronics Using Ensemble Learning

Majdi I. Radaideh, Chris Pappas, Mark Wezensky, Pradeep Ramuhalli, Sarah Cousineau (2022)

arXiv

Machine learning-based analysis of experimental electron beams and gamma energy distributions

M. Yadav, M. Oruganti, S. Zhang, B. Naranjo, G. Andonian, Y. Zhuang, Ö. Apsimon, C. P. Welsch, J. B. Rosenzweig (2023)

arXiv

Review of Time Series Forecasting Methods and Their Applications to Particle Accelerators

Sichen Li, Andreas Adelman (2022)

arXiv

Automatic setup of 18 MeV electron beamline using machine learning

Francesco Maria Velotti, Brennan Goddard, Verena Kain, Rebecca Ramjiawan, Giovanni Zevi Della Porta, Simon Hirlander (2022)

arXiv

Diagnostics for Linac Optimization With Machine Learning

R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2022)

arXiv

Transverse phase space tomography in the CLARA accelerator test facility using image compression and machine learning

Andrzej Wolski, Mark A. Johnson, Matthew King, Boris L. Militsyn, Peter H. Williams (2022)

arXiv

Using Kernel-Based Statistical Distance to Study the Dynamics of Charged Particle Beams in Particle-Based Simulation Codes

Chad E. Mitchell, Robert D. Ryne, Kilean Hwang (2022)

arXiv

Neural Network Solver for Coherent Synchrotron Radiation Wakefield Calculations in Accelerator-based Charged Particle Beams

Auralee Edelen, Christopher Mayes (2022)

arXiv

Adaptive Machine Learning for Time-Varying Systems: Towards 6D Phase Space Diagnostics of Short Intense Charged Particle Beams

Alexander Scheinker, Spencer Gessner (2022)

arXiv

Differentiable Preisach Modeling for Characterization and Optimization of Accelerator Systems with Hysteresis

R. Roussel, A. Edelen, D. Ratner, K. Dubey, J. P. Gonzalez-Aguilera, Y. K. Kim, N. Kuklev (2022)

arXiv

Mixed Diagnostics for Longitudinal Properties of Electron Bunches in a Free-Electron Laser

J. Zhu, N. M. Lockmann, M. K. Czwalińska, H. Schlarb (2022)

arXiv

A Neural Network Model of a Quasi-Periodic Elliptically Polarizing Undulator in Universal Mode

Ryan Sheppard, Cameron Baribeau, Tor Pedersen, Mark Boland, Drew Bertwistle (2022)

arXiv

Physics-informed neural network method for modelling beam-wall interactions

Kazuhiro Fujita (2022)

arXiv

Anomaly Detection in Particle Accelerators using Autoencoders

Jonathan P. Edelen, Nathan M. Cook (2021)

arXiv

Input Beam Matching and Beam Dynamics Design Optimization of the IsoDAR RFQ using Statistical and Machine Learning Techniques

Daniel Koser, Loyd Waites, Daniel Winklehner, Matthias Frey, Andreas Adelmann, Janet Conrad (2021)

arXiv

Neural Networks for ID Gap Orbit Distortion Compensation in PETRA III

Bianca Veglia, Ilya Agapov, Joachim Keil (2024)

arXiv

Time-Delayed Koopman Network-Based Model Predictive Control for the FRIB RFQ

Jinyu Wan, Shen Zhao, Wei Chang, Yue Hao (2024)

arXiv

Uncertainty Aware ML-based surrogate models for particle accelerators: A Study at the Fermilab Booster Accelerator Complex

Malachi Schram, Kishansingh Rajput, Karthik Somayaji Peng Li, Jason St. John, Himanshu Sharma (2022)

arXiv

Quantifying Uncertainty for Machine Learning Based Diagnostic

Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)

arXiv

Uncertainty Quantification for Virtual Diagnostic of Particle Accelerators

Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)

arXiv

Adaptive Latent Space Tuning for Non-Stationary Distributions

Alexander Scheinker, Frederick Cropp, Sergio Paiagua, Daniele Filippetto (2021)

arXiv

Adaptive deep learning for time-varying systems with hidden parameters: Predicting changing input beam distributions of compact particle accelerators

Alexander Scheinker, Frederick Cropp, Sergio Paiagua, Daniele Filippetto (2021)

arXiv

Using LSTM recurrent neural networks for monitoring the LHC superconducting magnets

Maciej Wielgosz, Andrzej Skoczko, Matej Mertik (2017)

arXiv

Coincident Learning for Beam-based RF Station Fault Identification Using Phase Information at the SLAC Linac Coherent Light Source

Jia Liang, William Colucho, Franz-Josef Decker, Ryan Humble, Ben Morris, Finn H. O'Shea, David A. Steele, Zhe Zhang, Eric Darve, Daniel Ratner (2025)

arXiv

Using Convolutional Neural Networks to Accelerate 3D Coherent Synchrotron Radiation Computations

Christopher Leon, Petr M. Anisimov, Nikolai Yampolsky, Alexander Scheinker (2025)

arXiv

Explainable physics-based constraints on reinforcement learning for accelerator controls

Jonathan Colen, Malachi Schram, Kishansingh Rajput, Armen Kasparian (2025)

arXiv

Accelerator system parameter estimation using variational autoencoded latent regression

Mahindra Rautela, Alan Williams, Alexander Scheinker (2024)

arXiv

Long Short-Term Memory Networks for Anomaly Detection in Magnet Power Supplies of Particle Accelerators

Ihar Lobach, Michael Borland (2024)

arXiv

Leveraging Prior Mean Models for Faster Bayesian Optimization of Particle Accelerators

Tobias Boltz, Jose L. Martinez, Connie Xu, Kathryn R. L. Baker, Zihan Zhu, Jenny Morgan, Ryan Roussel, Daniel Ratner, Brahim Mustapha, Auralee L. Edelen (2025)

arXiv

Optimizing Dynamic Aperture Studies with Active Learning

D. Di Croce, M. Giovannozzi, E. Krymova, T. Pieloni, S. Redaelli, M. Seidel, R. Tomás, F. F. Van der Veken (2024)

arXiv

Neural Network Prior Mean for Particle Accelerator Injector Tuning

Connie Xu, Ryan Roussel, Auralee Edelen (2022)

arXiv

Applications of object detection networks at high-power laser systems and experiments

Jinpu Lin, Florian Haberstroh, Stefan Karsch, Andreas Döpp (2022)

arXiv

A Neural Network approach to reconstructing SuperKEKB beam parameters from beamstrahlung

S. Di Carlo, G. Bonvicini, N. A. Althubiti, R. Ayad, E. De La Cruz-Burelo, I. Domínguez, B. O. El Bashir, H. Farhat, J. Flanagan, R. Gillard, S. Izaguirre Gamez, K. Kanazawa, K. Kumara, D. Liventsev, P. L. M. Podesta-Lerma, D. Ricalde-Herrmann, D. Rodriguez Perez, G. Tejeda-Muñoz,

M. Tobiyama I. Heredia de la Cruz (2022)
arXiv

Optimizing a Superconducting Radiofrequency Gun Using Deep Reinforcement Learning
David Meier, Luis Vera Ramirez, Jens Völker, Bernhard Sick, Jens Viefhaus, Gregor Hartmann
(2022)
arXiv

Uncertainty aware anomaly detection to predict errant beam pulses in the SNS accelerator
Willem Blokland, Pradeep Ramuhalli, Charles Peters, Yigit Yucesan, Alexander Zhukov, Malachi Schram, Kishansingh Rajput, Torri Jeske (2021)
arXiv

Adaptive Machine Learning for Time-Varying Systems: Low Dimensional Latent Space Tuning
Alexander Scheinker (2021)
arXiv

Fast, efficient and flexible particle accelerator optimisation using densely connected and invertible neural networks
Renato Bellotti, Romana Boiger, Andreas Adelmann (2021)
arXiv

Invertible Surrogate Models: Joint surrogate modelling and reconstruction of Laser-Wakefield Acceleration by invertible neural networks
Friedrich Bethke, Richard Pausch, Patrick Stiller, Alexander Debus, Michael Bussmann, Nico Hoffmann (2021)
arXiv

Improving Surrogate Model Accuracy for the LCLS-II Injector Frontend Using Convolutional Neural Networks and Transfer Learning
Lipi Gupta, Auralee Edelen, Nicole Neveu, Aashwin Mishra, Christopher Mayes, Young-Kee Kim (2021)
arXiv

A Novel Approach for Classification and Forecasting of Time Series in Particle Accelerators
Sichen Li, Mélissa Zacharias, Jochem Snuverink, Jaime Coello de Portugal, Fernando Perez-Cruz, Davide Reggiani, Andreas Adelmann (2021)
arXiv

Real-time Artificial Intelligence for Accelerator Control: A Study at the Fermilab Booster
Jason St. John, Christian Herwig, Diana Kafkes, Jovan Mitrevski, William A. Pellico, Gabriel N. Perdue, Andres Quintero-Parra, Brian A. Schupbach, Kiyomi Seiya, Nhan Tran, Malachi Schram, Javier M. Duarte, Yunzhi Huang, Rachael Keller (2021)
arXiv

Surrogate Modeling of the CLIC Final-Focus System using Artificial Neural Networks
J. Ogren, C. Gohil, D. Schulte (2021)
arXiv

Physics-Based Deep Neural Networks for Beam Dynamics in Charged Particle Accelerators
Andrei Ivanov, Ilya Agapov (2020)
arXiv

Introduction to Machine Learning for Accelerator Physics
Daniel Ratner (2020)
arXiv

Machine learning for design optimization of storage ring nonlinear dynamics
Faya Wang, Minghao Song, Auralee Edelen, Xiaobiao Huang (2019)
arXiv

Studies in Applying Machine Learning to LLRF and Resonance Control in Superconducting RF Cavities

Jorge Alberto Diaz Cruz, Sandra Biedron, Manel Martinez-Ramon, Salvador Sosa, Reza Pirayesh (2019)
arXiv

The model of an anomaly detector for HiLumi LHC magnets based on Recurrent Neural Networks and adaptive quantization

Maciej Wielgosz, Matej Mertik, Andrzej Skoczko, Ernesto De Matteis (2017)
arXiv

Machine learning for analysis of plasma driven ion source

N. Joshi, O. Meusel, H. Podlech (2018)
arXiv

First Steps Toward Incorporating Image Based Diagnostics Into Particle Accelerator Control Systems Using Convolutional Neural Networks

A. L. Edelen, S. G. Biedron, S. V. Milton, J. P. Edelen (2016)
arXiv

Neural Networks for Modeling and Control of Particle Accelerators

A. L. Edelen, S. G. Biedron, B. E. Chase, D. Edstrom, S. V. Milton, P. Stabile (2016)
arXiv

Harnessing the Power of Gradient-Based Simulations for Multi-Objective Optimization in Particle Accelerators

Kishansingh Rajput, Malachi Schram, Auralee Edelen, Jonathan Colen, Armen Kasparian, Ryan Roussel, Adam Carpenter, He Zhang, Jay Benesch (2024)
arXiv

BOOSTR: A Dataset for Accelerator Control Systems

Diana Kafkes, Jason St. John (2021)
arXiv

Model-free and Bayesian Ensembling Model-based Deep Reinforcement Learning for Particle Accelerator Control Demonstrated on the FERMI FEL

Simon Hirlaender, Niky Bruchon (2022)
arXiv

Autonomous Control of a Particle Accelerator using Deep Reinforcement Learning

Xiaoying Pang, Sunil Thulasidasan, Larry Rybarcyk (2020)
arXiv

AI-Assisted Transport of Radioactive Ion Beams

Sergio Lopez-Caceres, Daniel Santiago-Gonzalez (2025)
arXiv

Bayesian Optimization Algorithms for Accelerator Physics

Ryan Roussel, Auralee L. Edelen, Tobias Boltz, Dylan Kennedy, Zhe Zhang, Fuhao Ji, Xiaobiao Huang, Daniel Ratner, Andrea Santamaria Garcia, Chenran Xu, Jan Kaiser, Angel Ferran Pousa, Annika Eichler, Jannis O. Lubsen, Natalie M. Isenberg, Yuan Gao, Nikita Kuklev, Jose Martinez, Brahim Mustapha, Verena Kain, Weijian Lin, Simone Maria Liuzzo, Jason St. John, Matthew J. V. Streeter, Remi Lehe, Willie Neiswanger (2024)
arXiv

SPIRAL2 Cryomodules Models: a Gateway to Process Control and Machine Learning

Adrien Vassal, Adnan Ghribi, François Millet, François Bonne, Patrick Bonnay, Pierre-Emmanuel Bernaudin (2021)
arXiv

Predicting Beam Transmission Using 2-Dimensional Phase Space Projections Of Hadron Accelerators

Anthony Tran, Yue Hao, Brahim Mustapha, Jose L. Martinex Marin (2022)

arXiv

Machine Learning for Orders of Magnitude Speedup in Multi-Objective Optimization of Particle Accelerator Systems

Auralee Edelen, Nicole Neveu, Yannick Huber, Mattias Frey, Christopher Mayes, Andreas Adelman (2020)

arXiv

Machine learning assisted non-destructive transverse beam profile imaging

Zhanibek Omarov, Selcuk Haciomeroglu (2021)

arXiv

GAIA: A General AI Assistant for Intelligent Accelerator Operations

Frank Mayet (2024)

arXiv

AI-Enabled Operations at Fermi Complex: Multivariate Time Series Prediction for Outage Prediction and Diagnosis

Milan Jain, Burcu O. Mutlu, Caleb Stam, Jan Strube, Brian A. Schupbach, Jason M. St. John, William A. Pellico (2025)

arXiv

Automated GI tract segmentation using deep learning

Manhar Sharma (2023)

arXiv

Domain Adaptation of Automated Treatment Planning from Computed Tomography to Magnetic Resonance

Aly Khalifa, Jeff Winter, Inmaculada Navarro, Chris McIntosh, Thomas G. Purdie (2022)

arXiv

Superconducting radio-frequency cavity fault classification using machine learning at Jefferson Laboratory

Chris Tennant, Adam Carpenter, Tom Powers, Anna Shabalina Solopova, Lasitha Vidyaratne, Khan Iftekharuddin (2020)

arXiv

Generative Adversarial Networks (GAN) for compact beam source modelling in Monte Carlo simulations

David Sarrut, Nils Krah, Jean-Michel Létang (2019)

arXiv

Machine learning applied to single-shot x-ray diagnostics in an XFEL

A. Sanchez-Gonzalez, P. Micaelli, C. Olivier, T. R. Barillot, M. Ilchen, A. A. Lutman, A. Marinelli, T. Maxwell, A. Achner, M. Agåker, N. Berrah, C. Bostedt, J. Buck, P. H. Bucksbaum, S. Carron, B. Cooper, J. P. Cryan, M. Dong, R. Feifel, L. J. Frasinski, H. Fukuzawa, A. Galler, G. Hartmann, N. Hartmann, W. Helml, A. S. Johnson, A. Knie, A. O. Lindahl, J. Liu, K. Motomura, M. Mucke, C. O'Grady, J-E. Rubensson, E. R. Simpson, R. J. Squibb, C. Sâthe, K. Ueda, M. Vacher, D. J. Walke, V. Zhaunerchyk, R. N. Coffee, J. P. Marangos (2016)

arXiv

Action-Attentive Deep Reinforcement Learning for Autonomous Alignment of Beamlines

Siyu Wang, Shengran Dai, Jianhui Jiang, Shuang Wu, Yufei Peng, Junbin Zhang (2024)

arXiv

Image Segmentation using U-Net Architecture for Powder X-ray Diffraction Images

Howard Yanxon, Eric Roberts, Hannah Parraga, James Weng, Wenqian Xu, Uta Ruett, Alexander Hexemer, Petrus Zwart, Nickolas Schwarz (2023)

arXiv

Closing the loop: Autonomous experiments enabled by machine-learning-based online data analysis in synchrotron beamline environments

Linus Pithan, Vladimir Starostin, David Mareš, Lukas Petersdorf, Constantin Völter, Valentin Munteanu, Maciej Jankowski, Oleg Kononov, Alexander Gerlach, Alexander Hinderhofer, Bridget Murphy, Stefan Kowarik, Frank Schreiber (2023)
arXiv

Artifact Identification in X-ray Diffraction Data using Machine Learning Methods

Howard Yanxon, James Weng, Hannah Parraga, Wenqian Xu, Uta Ruett, Nicholas Schwarz (2022)
arXiv

An efficient search-and-score algorithm for ancestral graphs using multivariate information scores

Nikita Lagrange, Herve Isambert (2024)
arXiv

Loss function to optimise signal significance in particle physics

Jai Bardhan, Cyrin Neeraj, Subhadip Mitra, Tanumoy Mandal (2024)
arXiv

Efficient Compression of Sparse Accelerator Data Using Implicit Neural Representations and Importance Sampling

Xihaier Luo, Samuel Lurvey, Yi Huang, Yihui Ren, Jin Huang, Byung-Jun Yoon (2024)
arXiv

Variable Rate Neural Compression for Sparse Detector Data

Yi Huang, Yeonju Go, Jin Huang, Shuhang Li, Xihaier Luo, Thomas Marshall, Joseph Osborn, Christopher Pinkenburg, Yihui Ren, Evgeny Shulga, Shinjae Yoo, Byung-Jun Yoon (2024)
arXiv

Inverse Surrogate Model of a Soft X-Ray Spectrometer using Domain Adaptation

Enrico Ahlers, Peter Feuer-Forsen, Gregor Hartmann, Rolf Mitzner, Peter Baumgärtel, Jens Viehhaus (2025)
arXiv

Online tuning and light source control using a physics-informed Gaussian process

Adi Hanuka, J. Duris, J. Shtalenkova, D. Kennedy, A. Edelen, D. Ratner, X. Huang (2019)
arXiv

Deep-learning real-time phase retrieval of imperfect diffraction patterns from X-ray free-electron lasers

Sung Yun Lee, Do Hyung Cho, Chulho Jung, Daeho Sung, Daewoong Nam, Sangsoo Kim, Changyong Song (2024)
arXiv

A Two-Stage Machine Learning-Aided Approach for Quench Identification at the European XFEL

Lynda Boukela, Annika Eichler, Julien Branlard, Nur Zulaiha Jomhari (2024)
arXiv

Beam Detection Based on Machine Learning Algorithms

Haoyuan Li, Qing Yin (2023)
arXiv

Weakly supervised learning for pattern classification in serial femtosecond crystallography

Jianan Xie, Ji Liu, Chi Zhang, Xihui Chen, Ping Huai, Jie Zheng, Xiaofeng Zhang (2023)
arXiv

SpeckleNN: A unified embedding for real-time speckle pattern classification in X-ray single-particle imaging with limited labeled examples

Cong Wang, Eric Florin, Hsing-Yin Chang, Jana Thayer, Chun Hong Yoon (2023)
arXiv

Accurate and confident prediction of electron beam longitudinal properties using spectral virtual diagnostics

A. Hanuka, C. Emma, T. Maxwell, A. Fisher, B. Jacobson, M. J. Hogan, Z. Huang (2020)
arXiv

TempoRL: laser pulse temporal shape optimization with Deep Reinforcement Learning

Francesco Capuano, Davorin Peceli, Gabriele Tiboni, Raffaello Camoriano, Bedrich Rus (2023)
arXiv

A Start To End Machine Learning Approach To Maximize Scientific Throughput From The LCLS-II-HE

Aashwin Mishra, Matt Seaberg, Ryan Roussel, Fred Poitevin, Jana Thayer, Daniel Ratner, Auralee Edelen, Apurva Mehta (2025)
arXiv

Combined track finding with GNN & CKF

Lukas Heinrich, Benjamin Huth, Andreas Salzburger, Tilo Wettig (2024)
arXiv

Improved selective background Monte Carlo simulation at Belle II with graph attention networks and weighted events

Boyang Yu, Nikolai Hartmann, Luca Schinnerl, Thomas Kuhr (2023)
arXiv

Towards Reliable Neural Generative Modeling of Detectors

Lucio Anderlini, Matteo Barbetti, Denis Derkach, Nikita Kazeev, Artem Maevskiy, Sergei Mokhnenko (2022)
arXiv

Hybrid Quantum Classical Graph Neural Networks for Particle Track Reconstruction

Cenk Tüysüz, Carla Rieger, Kristiane Novotny, Bilge Demirköz, Daniel Dobos, Karolos Potamianos, Sofia Vallecorsa, Jean-Roch Vlimant, Richard Forster (2021)
arXiv

Graph Generative Models for Fast Detector Simulations in High Energy Physics

Ali Hariri, Darya Dyachkova, Sergei Gleyzer (2021)
arXiv

Beyond 4D Tracking: Using Cluster Shapes for Track Seeding

Patrick J. Fox, Shangqing Huang, Joshua Isaacson, Xiangyang Ju, Benjamin Nachman (2021)
arXiv

Fast Data-Driven Simulation of Cherenkov Detectors Using Generative Adversarial Networks

Artem Maevskiy, Denis Derkach, Nikita Kazeev, Andrey Ustyuzhanin, Maksim Artemev, Lucio Anderlini (2019)
arXiv

Accelerating the BSM interpretation of LHC data with machine learning

Gianfranco Bertone, Marc Peter Deisenroth, Jong Soo Kim, Sebastian Liem, Roberto Ruiz de Austri, Max Welling (2016)
arXiv

Radar Pulse Deinterleaving with Transformer Based Deep Metric Learning

Edward Gunn, Adam Hosford, Daniel Mannion, Jarrod Williams, Varun Chhabra, Victoria Nockles (2025)
arXiv

Deep Spectrum Cartography: Completing Radio Map Tensors Using Learned Neural Models

Sagar Shrestha, Xiao Fu, Mingyi Hong (2022)
arXiv

Radar Emitter Classification with Attribute-specific Recurrent Neural Networks

Paolo Notaro, Magdalini Paschali, Carsten Hopke, David Wittmann, Nassir Navab (2019)

arXiv

Prediction of Infinite Words with Automata

Tim Smith (2016)

arXiv

Learning topological defects formation with neural networks in a quantum phase transition

Han-Qing Shi, Hai-Qing Zhang (2023)

arXiv

A Conceptual Development of Quench Prediction App build on LSTM and ELQA framework

Matej Mertik, Maciej Wielgosz, Andrzej Skocze (2016)

arXiv

High-Precision Inversion of Dynamic Radiography Using Hydrodynamic Features

Maliha Hossain, Balasubramanya T. Nadiga, Oleg Korobkin, Marc L. Klasky, Jennifer L. Schei, Joshua W. Burby, Michael T. McCann, Trevor Wilcox, Soumi De, Charles A. Bouman (2021)

arXiv

Physical knowledge improves prediction of EM Fields

Andrzej Dulny, Farzad Jabbarigargari, Andreas Hotho, Laura Maria Schreiber, Maxim Terekhov, Anna Krause (2025)

arXiv

Plasma Surrogate Modelling using Fourier Neural Operators

Vignesh Gopakumar, Stanislas Pamela, Lorenzo Zanisi, Zongyi Li, Ander Gray, Daniel Brennand, Nitesh Bhatia, Gregory Stathopoulos, Matt Kusner, Marc Peter Deisenroth, Anima Anandkumar, JOREK Team, MAST Team (2024)

arXiv

AI-assisted Optimization of the ECCE Tracking System at the Electron Ion Collider

C. Fanelli, Z. Papandreou, K. Suresh, J. K. Adkins, Y. Akiba, A. Albataineh, M. Amaryan, I. C. Arsene, C. Ayerbe Gayoso, J. Bae, X. Bai, M. D. Baker, M. Bashkanov, R. Bellwied, F. Benmokhtar, V. Berdnikov, J. C. Bernauer, F. Bock, W. Boeglin, M. Borysova, E. Brash, P. Brindza, W. J. Briscoe, M. Brooks, S. Bueltmann, M. H. S. Bukhari, A. Bylinkin, R. Capobianco, W. -C. Chang, Y. Cheon, K. Chen, K. -F. Chen, K. -Y. Cheng, M. Chiu, T. Chujo, Z. Citron, E. Cline, E. Cohen, T. Cormier, Y. Corrales Morales, C. Cotton, J. Crafts, C. Crawford, S. Creekmore, C. Cuevas, J. Cunningham, G. David, C. T. Dean, M. Demarteau, S. Diehl, N. Doshita, R. Dupre, J. M. Durham, R. Dzhygadlo, R. Ehlers, L. El Fassi, A. Emmert, R. Ent, R. Fatemi, S. Fegan, M. Finger, M. Finger Jr., J. Frantz, M. Friedman, I. Friscic, D. Gangadharan, S. Gardner, K. Gates, F. Geurts, R. Gilman, D. Glazier, E. Glimos, Y. Goto, N. Grau, S. V. Greene, A. Q. Guo, L. Guo, S. K. Ha, J. Haggerty, T. Hayward, X. He, O. Hen, D. W. Higinbotham, M. Hoballah, T. Horn, A. Hognmrtsyan, P. -h. J. Hsu, J. Huang, G. Huber, A. Hutson, K. Y. Hwang, C. Hyde, M. Inaba, T. Iwata, H. S. Jo, K. Joo, N. Kalantarians, G. Kalicy, K. Kawade, S. J. D. Kay, A. Kim, B. Kim, C. Kim, M. Kim, Y. Kim, Y. Kim, E. Kistenev, V. Klimenko, S. H. Ko, I. Korover, W. Korsch, G. Krintiras, S. Kuhn, C. -M. Kuo, T. Kutz, J. Lajoie, D. Lawrence, S. Lebedev, H. Lee, J. S. H. Lee, S. W. Lee, Y. -J. Lee, W. Li, W. B. Li, X. Li, X. Li, X. Li, X. Li, Y. T. Liang, S. Lim, C. -h. Lin, D. X. Lin, K. Liu, M. X. Liu, K. Livingston, N. Liyanage, W. J. Llope, C. Loizides, E. Long, R. -S. Lu, Z. Lu, W. Lynch, D. Marchand, M. Marcisovsky, P. Markowitz, H. Marukyan, P. McGaughey, M. Mihovilovic, R. G. Milner, A. Milov, Y. Miyachi, A. Mkrtchyan, P. Monaghan, R. Montgomery, D. Morrison, A. Movsisyan, H. Mkrtchyan, A. Mkrtchyan, C. Munoz Camacho, M. Murray, K. Nagai, J. Nagle, I. Nakagawa, C. Nattrass, D. Nguyen, S. Niccolai, R. Nouicer, G. Nukazuka, M. Nycz, V. A. Okorokov, S. Oresic, J. D. Osborn, C. O'Shaughnessy, S. Paganis, S. F. Pate, M. Patel, C. Paus, G. Penman, M. G. Perdekamp, D. V. Perepelitsa, H. Periera da Costa, K. Peters, W. Phelps, E. Piasetzky, C. Pinkenburg, I. Prochazka, T. Protzman, M. L. Purschke, J. Putschke, J. R. Pybus, R. Rajput-Ghoshal, J. Rasson, B. Raue, K. F. Read, K. Roed, R. Reed, J. Reinhold, E. L. Renner, J. Richards, C. Riedl, T. Rinn, J. Roche, G. M. Roland, G. Ron, M. Rosati, C. Royon, J. Ryu, S. Salur, N. Santiesteban, R. Santos, M. Sarsour, J. Schambach, A. Schmidt, N. Schmidt, C. Schwarz, J. Schwiening, R. Seidl, A. Sickles, P. Simmerling, S. Sirca, D. Sharma, Z. Shi, T. -A. Shibata, C. -W. Shih, S. Shimizu, U. Shrestha, K.

Slifer, K. Smith, D. Sokhan, R. Soltz, W. Sondheim, J. Song, J. Song, I. I. Strakovsky, P. Steinberg, P. Stepanov, J. Stevens, J. Strube, P. Sun, X. Sun, V. Tadevosyan, W. -C. Tang, S. Tapia Araya, S. Tarafdar, L. Teodorescu, A. Timmins, L. Tomasek, N. Trotta, R. Trotta, T. S. Tveter, E. Umaka, A. Usman, H. W. van Hecke, C. Van Hulse, J. Velkovska, E. Voutier, P. K. Wang, Q. Wang, Y. Wang, Y. Wang, D. P. Watts, N. Wickramaarachchi, L. Weinstein, M. Williams, C. -P. Wong, L. Wood, M. H. Wood, C. Woody, B. Wyslouch, Z. Xiao, Y. Yamazaki, Y. Yang, Z. Ye, H. D. Yoo, M. Yurov, N. Zachariou, W. A. Zajc, W. Zha, J. Zhang, Y. Zhang, Y. X. Zhao, X. Zheng, P. Zhuang (2022)
arXiv

A Realistic Collimated X-Ray Image Simulation Pipeline

Benjamin El-Zein, Dominik Eckert, Thomas Weber, Maximilian Rohleder, Ludwig Ritschl, Steffen Kappler, Andreas Maier (2024)
arXiv

Online Electron Reconstruction at CLAS12

Tyson, Richard, Gavalian, Gagik (2025)
EPJ Web Conf.

Symplectic machine learning model for fast simulation of space-charge effects

Wan, Jinyu, Qiang, Ji, Hao, Yue (2025)
Phys.Rev.Accel.Beams

Application of quantum machine learning using variational quantum classifier in accelerator physics

Yin, He-Xing, Hu, Zhi-Yuan, Zeng, Huan-Huan, Guan, Jia-Bao, Wang, Ji-ke (2025)
arXiv

Laser wakefield electron acceleration simulation using physics-informed diffusion probabilistic models

Jech, Matej, Kovalenko, Alexander, Lazzarini, Carlo Maria, Grittani, Gabriele Maria (2025)
Proc.SPIE Int.Soc.Opt.Eng.

Geoff: The Generic Optimization Framework & Frontend for Particle Accelerator Controls

Madysa, Penelope, Appel, Sabrina, Kain, Verena, Schenk, Michael (2025)
SoftwareX

A Start To End Machine Learning Approach To Maximize Scientific Throughput From The LCLS-II-HE

Mishra, Aashwin, Seaberg, Matt, Roussel, Ryan, Poitevin, Fred, Thayer, Jana, Ratner, Daniel, Edelen, Auralee, Mehta, Apurva (2025)
arXiv

Using a neural network model to guide protection heater design in Nb₃Sn accelerator magnets

Bakrani Balani, Shahriar, Salmi, Tiina (2025)
Supercond.Sci.Technol.

Particle identification in the GlueX detector with machine learning

Habjan, Eric, Dube, Richard, McIntyre, James, Edo, Mezmur, Jones, Richard (2025)
arXiv

Prediction of Radio-quiet Gamma-Ray Pulsar Distances Using the Fundamental Plane Relation

Angüner, Ekrem O■uzhan (2025)
Astrophys.J.

Graph Neural Networks for particle tracking in NA62 Experiment

Plini, L., Tinti, G., Spadaro, T., Galasso, F. (2025)
Nuovo Cim.C

Reinforcement Learning for Charged Particle Beam Control to Minimize Injection Mismatch in Particle Accelerators

Balasoorya, Thilina, Yoo, Shinjae, Schoefer, Vincent, Tseng, Huan-Hsin, Gao, Yuan, Lin, Weijian, Silva, Chanaka De (2025)
Journal

Outlook towards deployable continual learning for particle accelerators

Rajput, Kishansingh, Lin, Sen, Edelen, Auralee, Blokland, Willem, Schram, Malachi (2025)
Mach.Learn.Sci.Tech.

An advanced pulse-avalanche stochastic model of long gamma-ray burst light curves

Maistrello, Manuele, Ferro, Lisa, Bazzanini, Lorenzo, Maccary, Romain, Guidorzi, Cristiano (2025)
Astron.Astrophys.

Application of ensemble machine learning algorithms and filtering techniques in slow orbit feedback systems of electron storage rings

Fan, Jiaqi, Liu, Weibin, Wang, Jiuqing, Wei, Yanru, Wei, Yuanyuan, Ji, Daheng (2025)
Phys.Rev.Accel.Beams

Integration of Machine Learning-Based Plasma Acceleration Simulations into Geant4: A Case Study with the PALLAS Experiment

Sytov, A., Cassou, K., Kubytskyi, V., Lenivenko, M., Huber, A. (2025)
arXiv

Optimisation of the Accelerator Control by Reinforcement Learning: A Simulation-Based Approach

Ibrahim, Anwar, Derkach, Denis, Petrenko, Alexey, Ratnikov, Fedor, Kaledin, Maxim (2025)
arXiv

Explainable physics-based constraints on reinforcement learning for accelerator controls

Colen, Jonathan, Schram, Malachi, Rajput, Kishansingh, Kasparian, Armen (2025)
arXiv

femto-PIXAR: a self-supervised neural network method for reconstructing femtosecond X-ray free electron laser pulses

Goetzke, Gesa, Plumley, Rajan, Hartmann, Gregor, Maxwell, Tim, Decker, Franz-Josef, Lutman, Alberto, Dunne, Mike, Ratner, Daniel, Turner, Joshua J. (2025)
Opt.Express

Deep Lie Map Networks: A novel Approach to Infer Nonlinear Synchrotron Optics from Beam Oscillations

Caliari, Conrad (2025)
Thesis

Reconstructing time-of-flight detector values of angular streaking using machine learning

Meier, David, Viefhaus, Jens, Hartmann, Gregor, Helml, Wolfram, Otto, Thorsten, Sick, Bernhard (2025)
Phys.Rev.Accel.Beams

Software Upgrades for the High-Luminosity LHC

Shope, David Richard (2025)
PoS

AI-Enabled Operations at Fermi Complex: Multivariate Time Series Prediction for Outage Prediction and Diagnosis

Jain, Milan, Mutlu, Burcu O., Stam, Caleb, Strube, Jan, Schupbach, Brian A., John, Jason M.St., Pellico, William A. (2025)
arXiv

Machine Learning Methods for Tau Lepton Identification and Search for the Supersymmetric Partner of the Tau Lepton Using CMS Run 2 Data

Shchedrolosiev, Mykyta (2025)
Thesis

Artificial intelligence for advancing particle accelerators

Ghribi, Adnan, Cassou, Kevin, Dalena, Barbara, Eichler, Annika, Guler, Hayg, Mistry, Andrew K., Oeftiger, Adrian, Shea, Thomas, Valentino, Gianluca, Welsch, Carsten P. (2025)
Europhys.News

Electron neutrino selection with deep neural networks for the ICARUS experiment

Koh, Dae Heun (2025)

Thesis

Real-Time event reconstruction for Nuclear Physics Experiments using Artificial Intelligence

Gavalian, Gagik (2025)

EPJ Web Conf.

Machine Learning for Optimized Polarization at Jefferson Lab

Jeske, Torri, Kasparian, Armen, Lawrence, David, Britton, Thomas, Schram, Malachi, Moran, Patrick, Fanelli, Cristiano, Guo, Jiawei, Jarvis, Naomi, Maxwell, James, Keith, Chris (2025)

EPJ Web Conf.

Applying the machine learning methods to determine the linear optics parameters in the ThomX collector ring

Klekots, D., Bezshyyko, O., Golinka-Bezshyyko, L., Kubytskyi, V., Chaikovska, I. (2024)

Nucl.Phys.Atom.Energy

Deep learning framework for fault detection in accelerators

Piekarski, Michal (2024)

JACoW

Optimizing Beam-Plasma Interactions Through Jitter Analysis Using Start-to-End Simulations

Hwang, Robin (2024)

arXiv

Machine Learning Applications for Improving Accelerator Operations

Lin, Lucy (2024)

Thesis

Virtual pulse reconstruction diagnostic for single-shot measurement of free electron laser radiation power

Korten, Till, Rybnikov, Vladimir, Steinbach, Peter, Mirian, Najmeh (2024)

Phys.Rev.Accel.Beams

Assessing the Performance of Deep Learning Predictions for Dynamic Aperture of a Hadron Circular Particle Accelerator

Di Croce, Davide, Giovannozzi, Massimo, Montanari, Carlo Emilio, Pieloni, Tatiana, Redaelli, Stefano, Van der Veken, Frederik F. (2024)

Instruments

Machine learning

Snuverink, Jochem (2024)

CERN Yellow Rep.School Proc.

Harnessing Machine Learning for Single-Shot Measurement of Free Electron Laser Pulse Power

Korten, Till, Rybnikov, Vladimir, Vogt, Mathias, Roensch-Schulenburg, Juliane, Steinbach, Peter, Mirian, Najmeh (2024)

Journal

Data-driven gradient optimization for field emission management in a superconducting radio-frequency linac

Goldenberg, Steven, Ahammed, Kawser, Carpenter, Adam, Li, Jiang, Suleiman, Riad, Tennant, Chris (2024)

Phys.Rev.Accel.Beams

Third-integer Resonant Extraction Regulation System for Mu2e

Narayanan, Aakaash (2024)

InspireHEP

Harnessing the power of gradient-based simulations for multi-objective optimization in particle accelerators

Rajput, Kishansingh, Schram, Malachi, Edelen, Auralee, Colen, Jonathan, Kasparian, Armen, Roussel, Ryan, Carpenter, Adam, Zhang, He, Benesch, Jay (2024)

Mach.Learn.Sci.Tech.

Machine learning-based non-destructive measurement of bunch length at FRIB

Wan, Jinyu, Plastun, Alexander, Ostroumov, Peter (2024)

JACoW

Beam emittance and Twiss parameters from pepper-pot images using physically informed neural nets

Knight, Ian, Mustapha, Brahim (2024)

JACoW

Advancements in backwards differentiable beam dynamics simulations for accelerator design, model calibration, and machine learning

Roussel, Ryan, Edelen, Auralee, Gonzalez-Aguilera, Juan Pablo, Lehe, Remi, Huebl, Axel, Kaiser, Jan, Santamaria Garcia, Andrea, Xu, Chenran, Eichler, Annika, Charleux, Grégoire (2024)

JACoW

Advanced algorithms for linear accelerator design and operation

Ong, Ysabella Kassandra, Bellan, Luca, Pisent, Andrea, Comunian, Michele, Fagotti, Enrico, Bortolato, Damiano, Montis, Maurizio, Giacchini, Mauro, Carletto, Osvaldo (2024)

JACoW

Adaptive machine learning with hard physics constraints and generative diffusion for 6D phase space diagnostics

Scheinker, Alexander (2024)

JACoW

Data-Driven Discovery of Beam Centroid Dynamics

Pocher, Liam A., Haber, Irving, Antonsen, Thomas M., O'Shea, Patrick G. (2024)

arXiv

Signal model parameter scan using Normalizing Flow

Saito, Masahiko, Morinaga, Masahiro, Kishimoto, Tomoe, Tanaka, Junichi (2024)

PoS

Towards Agentic AI on Particle Accelerators

Sulc, Antonin, Hellert, Thorsten, Kammering, Raimund, Houscher, Hayden, St. John, Jason (2024)

Journal

Machine learning for reducing noise in RF control signals at industrial accelerators

Henderson, M., Edelen, J.P., Einstein-Curtis, J., Hall, C.C., Cruz, J.A. Diaz, Edelen, A.L. (2024)

JINST

Continuous data-driven control of the GTS-LHC ion source at CERN

Kain, Verena, Rodriguez Mateos, Borja, Bruchon, Niky, Hirlander, Simon, Kuchler, Detlef (2024)

JACoW

Surrogate Models studies for laser-plasma accelerator electron source design through numerical optimisation

Kane, G., Drobniak, P., Kazamias, S., Kubytskyi, V., Lenivenko, M., Lucas, B., Serhal, J., Cassou, K., Beck, A., Specka, A., Massimo, F. (2024)

arXiv

Beamline Steering Using Deep Learning Models

Allen, Dexter, Kante, Isaac, Bohler, Dorian (2024)

arXiv

Beam-based identification of magnetic field errors in a synchrotron using deep Lie map networks

Caliari, Conrad, Oeftiger, Adrian, Boine-Frankenheim, Oliver (2024)
Phys.Rev.Accel.Beams

Towards efficient machine-learning-based reduction of the cosmic-ray induced background in X-ray imaging detectors: increasing context awareness

Poliszczuk, Artem, Wilkins, Dan, Allen, Steven W., Miller, Eric D., Chattopadhyay, Tanmoy, Schneider, Benjamin, Darve, Julien Eric, Bautz, Marshall, Falcone, Abe, Foster, Richard, Grant, Catherine E., Herrmann, Sven, Kraft, Ralph, Morris, R. Glenn, Nulsen, Paul, Orel, Peter, Schellenberger, Gerrit, Stueber, Haley R. (2024)
Proc.SPIE Int.Soc.Opt.Eng.

A Two-Stage Machine Learning-Aided Approach for Quench Identification at the European XFEL

Boukela, Lynda, Eichler, Annika, Branlard, Julien, Jomhari, Nur Zulaiha (2024)
arXiv

Reinforcement learning-trained optimisers and Bayesian optimisation for online particle accelerator tuning

Kaiser, Jan, Xu, Chenran, Eichler, Annika, Santamaria Garcia, Andrea, Stein, Oliver, Bründermann, Erik, Kuropka, Willi, Dinter, Hannes, Mayet, Frank, Vinatier, Thomas, Burkart, Florian, Schlarb, Holger (2024)
Sci.Rep.

Machine learning-based extraction of longitudinal beam parameters in the LHC

Iliakis, Konstantinos, Karlsen-Bæck, Birk Emil, Trad, Georges, Timko, Helga, Zampetakis, Michail, Argyropoulos, Theodoros (2024)
JACoW

High fidelity numerical modelling and condition monitoring applied to septum magnets at CERN

Kawa, Krzysztof, Szumlak, Tomasz, Kokkinos, Charilaos, Lackner, Friedrich (2024)
JACoW

Updates to Xopt for online accelerator optimization and control

Roussel, Ryan, Kennedy, Dylan, Boltz, Tobias, Baker, Kathryn, Mayes, Christopher, Edelen, Auralee (2024)
JACoW

Breaking new ground in data-intensive science: first insights from the LIV.INNO center for doctoral training

Welsch, Carsten (2024)
JACoW

The reinforcement learning for autonomous accelerators collaboration

Santamaria Garcia, Andrea, Eichler, Annika, Xu, Chenran, Kaiser, Jan, Scomparin, Luca, Schenk, Michael, Pochaba, Sabrina, Hirlander, Simon (2024)
JACoW

Machine learning-based particle accelerator modeling

Emmanuel Goutierre (2024)
Laboratoire Interdisciplinaire des Sciences du Numérique

Modelization of an Injector With Machine Learning

Mathieu Debongnie, Maud Baylac, Frédéric Bouly, Nicolas Chauvin, Angélique Gatera, Tomas Junquera, Didier Uriot (2019)
10th International Particle Accelerator Conference

Field-Reliability Predictions Based on Statistical System Lifecycle Models

Lukas Felsberger, Dieter Kranzlmüller, Benjamin Todd (2018)

2nd International Cross-Domain Conference for Machine Learning and Knowledge Extraction (CD-MAKE)

A Machine Learning Technique for Dynamic Aperture Computation

Barbara Dalena, Mehdi Ben Ghali (2021)

12th International Particle Accelerator Conference

Surrogate Model for Linear Accelerator: A fast Neural Network approximation of ThomX's simulator

Emmanuel Goutierre, Christelle Bruni, Johanne Cohen, Hayg Guler, Michèle Sebag (2023)

IPAC 2023 - 14th International Particle Accelerator Conference

Physics-aware modelling of an accelerated particle cloud

Emmanuel Goutierre, Christelle Bruni, Johanne Cohen, Hayg Guler, Michèle Sebag (2023)

MLPS 2023 - Machine Learning and the Physical Sciences Workshop 23023 - At the 37th conference on Neural Information Processing Systems (NeurIPS)

Closing the loop: Autonomous experiments enabled by machine-learning-based online data analysis in synchrotron beamline environments

Linus Pithan, Vladimir Starostin, David Mareš, Lukas Petersdorf, Constantin Völter, Valentin Munteanu, Maciej Jankowski, Oleg Kononov, Alexander Gerlach, Alexander Hinderhofer, Bridget Murphy, Stefan Kowarik, Frank Schreiber (2023)

HAL

Leveraging Subgraph Extraction for Performance Portable Programming Frameworks on DL Accelerators

Xiao Zhang, Huiying Lan, Tian Zhi (2018)

15th IFIP International Conference on Network and Parallel Computing (NPC)

From Compact Plasma Particle Sources to Advanced Accelerators with Modeling at Exascale

Axel Huebl, Remi Lehe, Edoardo Zoni, Olga Shapoval, Ryan T Sandberg, Marco Garten, Arianna Formenti, Revathi Jambunathan, Prabhat Kumar, Kevin Gott, Andrew Myers, Weiqun Zhang, Ann Almgren, Chad E Mitchell, Ji Qiang, David Grote, Alexander Sinn, Severin Diederichs, Maxence Thevenet, Luca Fedeli, Thomas Clark, Neil Zaim, Henri Vincenti, Jean-Luc Vay (2022)

20th Advanced Accelerator Concepts Workshop (AAC'22)

3D deep convolutional neural network segmentation model for precipitate and porosity identification in synchrotron X-ray tomograms

S. Gaudez, M. Ben Haj Slama, A. Kaestner, Manas Vijay Upadhyay (2022)

Journal of Synchrotron Radiation

Leveraging Serial MRI Radiomics and Machine Learning to Predict Risk of Radiation Necrosis in Patients with Brain Metastases Managed with Stereotactic Radiation and Immunotherapy

H. Elhalawani, L.A. Hammoudeh, D.N. Cagney, J.M. Qian, A. Martin, J. Zgrabik, J. Meyers, K.J. Pataki, K. Martin, Y. Khouj, C. Verry, W.L. Bi, O. Arnaout, S.M. Christ, B. Alexander, S. Tanguturi, R. Rahman, D. Haas-Kogan, A.A. Aizer (2022)

64th ASTRO Annual Meeting

Neural networks for rapid phase quantification of cultural heritage X-ray powder diffraction data

Victor Poline, Ravi Raj Purohit, Purushottam Raj Purohit, Pierre Bordet, Nils Blanc, Pauline Martinetto (2024)

Journal of Applied Crystallography

Understanding error propagation in deep learning neural network (DNN) accelerators and applications

Guanpeng Li, Siva Kumar Sastry Hari, Michael B. Sullivan, Timothy Tsai, Karthik Pattabiraman, Joel Emer, Stephen W. Keckler (2017)

Unknown Venue

Survey and Benchmarking of Machine Learning Accelerators

Albert Reuther, Peter Michaleas, Michael Jones, Vijay Gadepally, Siddharth Samsi, Jeremy Kepner (2019)

Unknown Venue

A Survey of Accelerator Architectures for Deep Neural Networks

Yiran Chen, Yuan Xie, Linghao Song, Fan Chen, Tianqi Tang (2020)

Engineering

RBF neural net based classifier for the AIRIX accelerator fault diagnosis

J. C. Ribes, G. Delaunay, J. Delvaux, E. Merle, M. Mouillet (2000)

arXiv

Railway Wheel Impact Force and Alert Prediction Using Machine Learning Models

Gajendra Malviya, Shripad Salsingkar (2025)

AIRO Springer Series

RadiSimCLIP: A Radiology Vision-Language Model Pretrained on Simulated Radiologist Learning Dataset for Zero-Shot Medical Image Understanding

Minhui Tan, Qingxia Wu, Boyang Zhang, Genqiang Ren, Jianlong Nie, Zhong Xue, Xiaohuan Cao, Dinggang Shen (2025)

Lecture Notes in Computer Science

Machine Learning for Precision Dental Diagnosis: Real-Time X-Ray Detection and Patient Record Integration—SmiloScope

M. S. Padmini, Asha Rani Mahadeva, Madhu Nagaraj, Namratha Gopinath (2025)

Information Systems Engineering and Management

A Living Review Pipeline for AI/ML Applications in Accelerator Physics

Adnan Ghribi (2025)

arXiv

A Living Review Pipeline for AI/ML Applications in Accelerator Physics

Ghribi, Adnan (2025)

arXiv

Preventive Maintenance of Mining Excavators Using Machine Learning: Enhancing Operational Efficiency and Cost-Effectiveness

Syed Adnan Ahmad, Prakash Kumar (2025)

Lecture Notes on Multidisciplinary Industrial Engineering

Steel Surface Defects Recognition Based on Multi-Type Statistical Features and Multiclass Support Vector Machine

Dibya Tripathi, Kheelraj Pandey, A. K. Chauhan (2025)

Lecture Notes on Multidisciplinary Industrial Engineering

Prediction of Tensile Strength and Impact Strength in Fused Deposition Modeling Using a Machine Learning Pipeline

Anne Vogler, Benjamin Küster, Malte Stonis, Ludger Overmeyer (2025)

Lecture Notes in Mechanical Engineering

TAIDL: Tensor Accelerator ISA Definition Language with Auto-generation of Scalable Test Oracles

Devansh Jain, Marco Frigo, Jai Arora, Akash Pardeshi, Zhihao Wang, Krut Patel, Charith Mendis (2025)

Proceedings of the 2025 58th IEEE/ACM International Symposium on Microarchitecture

BitL: A Hybrid Bit-Serial and Parallel Deep Learning Accelerator for Critical Path Reduction

Seunghyun Lee, Dongho Ha, Sungbin Kim, Sungwoo Kim, Hyunwuk Lee, Won Woo Ro (2025)

Proceedings of the 2025 58th IEEE/ACM International Symposium on Microarchitecture

Bolt Anchorage Defect Identification Based on Ultrasonic Guided Wave and Deep Learning

Hui Xing, Weiguo Di, Xiaoyun Sun, Mingming Wang, Chaobo Li (2025)

Sensors

PSVM-MR: A Parallel Support Vector Machine Algorithm Based on MapReduce

Bin-bin Guo, Yimin Mao, A Yaser, Neelakandan Chandrasekaran, Le Kang, Wenhao Li, Decheng Miao (2025)

Communications in Computer and Information Science

DEEP LEARNING-BASED CLASSIFICATION OF SEISMIC DAMAGE ON PARTITION WALL AND ARTIFICIAL CRACKING PATTERN

Kotoka FUJISAWA, Noriyuki TAKAHASHI (2025)

AIJ Journal of Technology and Design

Enhancing RFI management in construction through machine learning-driven predictive models

Neziha Yilmaz, Esin Ergen, Ahmet Muhtar Citipitioglu (2025)

Smart and Sustainable Built Environment

Digital Twin Framework for PIP-II Linac: AI-Driven Multi-Scale Modeling from Ion Source to 800 MeV

Pathak, Abhishek, Hanlet, Pierrick, Miceli, Tia (2025)

Journal

A “Bottom-Up” Approach for Constructing Underground Engineering Knowledge Graph

Yiming Luo, Yunfei Xiang, Peng Lin, Yong Xia, Yuanguang Liu, Yao Xu, Chaoyi Li (2025)

Mechanisms and Machine Science

Achievement of High■Quality Gallium Oxide Epitaxial Growth via Machine Learning

Yaoping Lu, Yu Zhang, Ben Niu, Titao Li, Zhenjie Zheng, Lemin Jia, Duanyang Chen, Hongji Qi, Kelvin H. L. Zhang, Min Zhu, Haizhong Zhang, Xiaoqiang Lu (2025)

Advanced Functional Materials

Unsupervised Online Learning for AC Optimal Power Flow: A Gradient-Guided Physics-Informed Neural Network Approach

Bozhen Jiang, Jing Qu, Qin Wang (2025)

Institute of Electrical and Electronics Engineers (IEEE)

Enhancing Predictive Accuracy in Shear Strength of RC Deep Beams: A Comprehensive Analysis Using Ensemble Machine Learning Models

Arslan Qayyum Khan, Muhammad Huzaifa Naveed, Muhammad Dawood Rasheed, Amorn Pimanmas (2025)

Arabian Journal for Science and Engineering

Two-Step Deep Learning Approach for Fine-Grained Railway Catenary Segmentation

Milo Belien, Metehan Doyran (2025)

Proceedings of the International Workshop on Application-driven Point Cloud Processing and 3D Vision

Saffron Adulteration Prediction Based on Fine-Grained Deep Features

Ishrat Nazeer, Ranjeet Kumar Rout, Saiyed Umer (2025)

Lecture Notes in Networks and Systems

Machine-Learning Approach for Pulsed Electromagnetic Field Therapy Parameters Optimization for Enhanced Tissue Penetration

Pooja Kumari Jha, Vikas Kumar, Manoj Kumar Parida, S. Kanagaraj (2025)

Lecture Notes in Networks and Systems

Artificial Intelligence and Machine Learning Based Predictive Modelling of Beam and 3D Bolt Through FEA

Neha M. Deshmukh, Yashwant S. Munde, Surajit S. Wadagaonkar, Avinash S. Shinde, Prashant R. Anerao (2025)

Lecture Notes in Mechanical Engineering

Fault Prediction of 5G Base Station RF Module Based on Machine Learning Algorithms

Mingzan Ning (2025)

Applied and Computational Engineering

MiMu: mitigating multiple shortcut learning behavior of transformers

Lili Zhao, Qi Liu, Wei Chen, Liyi Chen, Ruijun Sun, Min Hou, Yang Wang, Shijin Wang, Pingping Ren, Jiafeng Zhou (2025)

Frontiers of Computer Science

Collective User Behavior Driven Self Learning Search Engine

Zhiar Piroti (2025)

Springer Science and Business Media LLC

Reinforcement Learning for Accelerator Beamline Control: a simulation-based approach

Anwar Ibrahim, Alexey Petrenko, Maxim Kaledin, Ehab Suleiman, Fedor Ratnikov, Denis Derkach (2025)

arXiv

Reinforcement Learning for Accelerator Beamline Control: a simulation-based approach

Ibrahim, Anwar, Petrenko, Alexey, Kaledin, Maxim, Suleiman, Ehab, Ratnikov, Fedor, Derkach, Denis (2025)

arXiv

Automated beam tuning of the TRIUMF ISAC facility

Shelbaya, Olivier, Jung, Paul, Kester, Oliver, Hassan, Omar (2025)

JACoW

Deep Learning-Based Classification of Spine X-Ray Images Using Attention Mechanisms

Asaram Pandurang Janwale, Minal Dutta, Savita Mohurle, Vaduguru Venkata Ramya (2025)

Lecture Notes in Networks and Systems

Abstract 4345868: Machine learning prediction of myocardial ischemia using quantitative assessment of coronary calcifications in non-contrast CT calcium scoring scans

Juhwan Lee, Mohamed H.E. Makhoul, Sadeer Al-Kindi, Ammar Hoori, Tao Hu, Hao Wu, Justin Kim, Sanjay Rajagopalan, David Wilson (2025)

Circulation

A data-driven hybrid method combining experiments, finite element modeling and machine learning for impact response prediction of TPU composites

Shunqi Zhang, Luca Lomazzi, Dayou Ma, Andrea Manes (2025)

International Journal of Structural Integrity

Human-in-the-loop reinforcement learning for data quality monitoring in particle physics experiments

Olivia Jullian Parra, Julián García Pardiñas, Lorenzo Del Pianta Pérez, Maximilian Janisch, Suzanne Klaver, Thomas Lehericy, Nicola Serra (2025)

Machine Learning: Science and Technology

ESTIMATIVA DE VIDA ÚTIL REMANESCENTE DE PLACAS CERÂMICAS UTILIZADAS EM FILTRAGEM DE MINÉRIO DE FERRO ATRAVÉS DE MODELO DE CONFIABILIDADE E MÉTODOS DE MACHINE LEARNING APLICADOS A VARIÁVEIS DE PROCESSO INDUSTRIAL DE UM SISTEMA PIMS

ROBERT BENTO FLORENTINO, LUIZ GUSTAVO LOURENÇO MOURA (2025)

Anais do Encontro Nacional de Engenharia de Produção

Reinforcement Learning for Accelerator Beamline Control: a simulation-based approach

Ibrahim Anwar, Petrenko, Alexey, Kaledin, Maxim, Suleiman, Ehab, Ratnikov, Fedor, Derkach Denis (2025)

arXiv (Cornell University)

A Machine Learning Approach to Li-Ion Battery Module

Ram U. Kalyankar, Kalpak R. Sagar, Vipul M. Patel, Jyotirmay Banerjee, Hemantkumar B. Mehta

(2025)

Lecture Notes in Mechanical Engineering

An Overview of Artificial Intelligence and Machine Learning

Adwika Singh, Falguni Sharma, Anshuman Shastri (2025)

Artificial Intelligence for Biomass-based Biofuel Production

IGD-YOLOv8s: insulator defect detection via iterative attention and generalized dynamic feature pyramids

Zhiqin Zhang, Guowu Yuan, Junlin Rao, Xiang Li, Pengfei Yu, Hao Zhou (2025)

Measurement Science and Technology

A Teachable Machine for Transformers

Nicolas Pope, Matti Tedre (2025)

Proceedings of the 25th Koli Calling International Conference on Computing Education Research

Predictive Modeling of ICU Mortality Using Supervised Machine Learning Algorithms

Malithi Upeksha, Sulanie Perera (2025)

Communications in Computer and Information Science

SABER: Symbolic Regression-based Angle of Arrival and Beam Pattern Estimator

Shih-Kai Chou, Mengran Zhao, Cheng-Nan Hu, Kuang-Chung Chou, Carolina Fortuna, Jernej Hribar (2025)

arXiv

Machine learning-driven optimization of energy absorption in PLA composites reinforced with carbon fibers through FFF additive manufacturing

Rezgar Hasanzadeh (2025)

Rapid Prototyping Journal

Beam Energy Forecasting using Machine Learning at the CLEAR accelerator

Gilardi, Antonio, Malyzhenkov, Alexander, Petersson, Alfred, Mostacci Andrea, Pollastro, Andrea, Aksoy, Avni, Filippetto, Daniele, Gamba, Davide, Granados, Eduardo, Tangari, Giacomo, Sjobak, Kyrre, Bonnard, Ladislav, Wroe, Laurence, Carranza-García, Manuel, Franek Ondrej, Korysko, Pierre, Corsini Roberto, Rieker, Vilde, Farabolini, Wilfrid, Mazzoni, Stefano (2025)

JACOW

Advancing accelerator science through data-intensive research and training

Welsch, Carsten (2025)

JACOW

Machine learning approach to MDI optimization for 3 TeV c.o.m. Muon Collider

Castelli, Luca (2025)

JACOW

Advances in machine learning inference of dynamic aperture evaluation for the LHC

Montanari, Carlo Emilio, Di Croce, Davide, Van der Veken, Frederik, Giovannozzi, Massimo, Appleby, Robert, Redaelli, Stefano, Pieloni, Tatiana (2025)

JACOW

Machine learning for calibration drift forecasting in superconducting RF cavities

Sun Yue, Bellandi Andrea, Eichler, Annika, Richter, Bozo, Diomedede, Marco, Schmidt, Christian, Schlarb Holger, Branlard, Julien, Herrmann, Max (2025)

JACOW

Uncertainty-Quantified Machine Model Construction Using Physics-Informed Gaussian Processes and Bayesian Optimization

Isensee, Victoria, Oeftiger, Adrian, Boine-Frankenheim, Oliver (2025)

JACOW

Physics-Informed Deep Learning for Improved Input Function Estimation in Motion-Blurred Dynamic ^{18}F FDG PET Images

Christian Salomonsen, Kristoffer K. Wickstrøm, Samuel Kuttner, Elisabeth Wetzer (2025)

Lecture Notes in Computer Science

Fusion model combining ultrasound-based radiomics and deep transfer learning with clinical parameters for preoperative prediction of pelvic lymph node metastasis in cervical cancer

Jihan Wang, Shengxian Bao, Tongtong Huang, Yongzhi Cai, Binbin Jin, Ji Wu (2025)

Frontiers in Oncology

Hybrid Machine Learning based Optimization with Beamforming of 6G Mobile Radio Network for LEO Satellite base station in Urban Areas

Haymanot Derebe Bizuneh (2025)

Springer Science and Business Media LLC

Fluence Map Prediction with Deep Learning: A Transformer-based Approach

Ujunwa Mgbob, Rafi Sultan, Dongxiao Zhu, Joshua Kim (2025)

arXiv

Machine Learning in High-Momentum Particle Identification in the MPD Experiment

Papoyan, V., Aparin, A., Ayriyan, A., Grigorian, H., Korobitsin, A. (2025)

Phys.Part.Nucl.

ChatEED: An agentic retrieval assistant for accelerator operators

Aaron Zachary Reed, Claudio Bisegni, Sandesh Shrestha, Michelle Huang, Daniel Ratner (2025)

Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis

Machine Learning in Semiconductor Failure Analysis: Techniques and Case Studies (2025 Update)

Michael Kögel, Sebastian Brand, Frank Altmann (2025)

International Symposium for Testing and Failure Analysis

NNia-8: An 8-Core RISC-V Neural Network Inference Accelerator with Efficient Processing Elements and Memory Utilization

Xingbo Wang, Yucong Huang, Xinyu Kang, Yuru Li, Qi Wang, Terry Tao Ye (2025)

Lecture Notes in Computer Science

Railway Track Obstacle Detection and Alert System Using Deep Learning

R. Anand, D. P. Monish, D. Madhukar, C. G. Paari (2025)

Lecture Notes in Networks and Systems

Deep Learning Assisted Modeling for χ^2 Nonlinear Optics

Jack Hirschman, Erfan Abedi, Minyang Wang, Hao Zhang, Abhimanyu Borthakur, Justin Baker, Andrea L. Bertozzi, Randy Lemons, Sergio Carbajo (2025)

arXiv

Robust Training to Secure Automated AI Accelerator Generation Against Malicious Platforms

Chao Guo, Youhua Shi (2025)

Communications in Computer and Information Science

Machine learning with ultrasound examination for prediction of intraoperative hypotension during robot-assisted radical prostatectomy

V. S. Andreenkov, A. V. Vlasenko, A. N. Kornienko, A. S. Kazakov, E. P. Rodionov, K. B.

Kolontarev, E. A. Evdokimov (2025)

Medical alphabet

Categories of Defect Prediction

Zhou Xu (2025)

Machine-Learning-Assisted Software Defect Prediction

Optimized Supervised Machine Learning for Accurate Estimation of Reinforcement in RC Beams and Columns

Nhan Thanh Vu Nguyen, Chon Tran, Duong Thai Le, Quy Thue Nguyen (2025)
Lecture Notes in Civil Engineering

Radial Pulse Pattern Recognition Using Deep Learning

Nishant Patil, Ghanshyam Jindal, Sanjeev Rai, Gajanan Nagare (2025)
Lecture Notes in Networks and Systems

Measurement and Evaluation of Intelligent Robots

Claude Sammut (2025)
Lecture Notes in Computer Science

Parameter estimation of three-phase voltage source converter using physics-informed machine learning

Cuiyu Liu, Zhiming Yang, Yipeng Yang, Ziwen Xiao, Chuanfeng Li, Yang Yu (2025)
Measurement Science and Technology

Automated RF Phase Adjustment for Beam Stabilization in the Fermilab Linac

R. R. Chichili, J. A. Sulskis, R. Sharankova, B. Vamanan, S. Ravi (2025)
arXiv

Fast Adaptive Neural Control of Resonant Extraction at Fermilab

A. Whitbeck, J. Berlioz, K. Danison-Fieldhouse, K. Hazelwood, M. Khan, J. Mitrevski, A. Narayanan, J. St. John, N. Tran, J. Ji, M. Walter (2025)
arXiv

Hybrid deep semantic query expansion using multi-objective fireworks-transformer optimization: a reinforcement learning approach

Fouad Bekkari, Med Redouane Kafi, Amine Khaldi (2025)
Evolutionary Intelligence

Automated RF Phase Adjustment for Beam Stabilization in the Fermilab Linac

Chichili, R.R., Sulskis, J.A., Sharankova, R., Vamanan, B., Ravi, S. (2025)
Journal

Fast Adaptive Neural Control of Resonant Extraction at Fermilab

Whitbeck, A., Berlioz, J., Danison-Fieldhouse, K., Hazelwood, K., Khan, M., Mitrevski, J., Narayanan, A., John, J.St., Tran, N., Ji, J., Walter, M. (2025)
Journal

Study on estimation of internal cracks using electromagnetic waves and machine learning

Hinata Yamamoto, Hitoshi Ito, Toshiaki Mizobuchi (2025)
Proceedings of the Institution of Civil Engineers - Construction Materials

A New Pulsed Neutron-Gamma Density Logging Method Based on Gamma-Ray Spectra and Machine Learning

Duo Dong, Qiong Zhang, Defu Zang, Wensheng Wu, He Meng, Yunlong Ge, Hu Wang, Xiaoyu Song, Shitao Xiong (2025)
ACS Omega

Figure S3 from Building Practical Risk Prediction Models for Nasopharyngeal Carcinoma Screening with Patient Graph Analysis and Machine Learning

Anjun Chen, Roufeng Lu, Ruobing Han, Ran Huang, Guanjie Qin, Jian Wen, Qinghua Li, Zhiyong Zhang, Wei Jiang (2025)
American Association for Cancer Research (AACR)

Innovative Approaches for Learning Electrical Concepts in Physics

Papun Mondal, Sanjoy Kumar Pal, Pradipta Panchadhyayee, Suman Kumar Dash, Subhash Chandra Samanta (2025)
Resonance

A study on sex prediction by using machine algorithms with anthropometric measurements of the seventh cervical vertebra

Esra Cetin Unlu, Zulal Oner, Serkan Oner, Muhammed Kamil Turan, Rukiye Sumeyye Bakici (2025)

Anthropologischer Anzeiger

Machine learning model for predicting the magnitude of tumor movement based on clinical information and CT-based radiomics features

Xinyu Song, Lian Duan, Guangyu Wang, Xiangyu Zhang, Qing Xiao, Guyu Dai, Jing Li, Yinglin Peng, Sen Bai, Guangjun Li (2025)

Applied Radiation and Isotopes

Integrated engineering framework for fatigue damage prediction of fighter aircraft using machine learning

Zahid Ahmad, Ghulam Murtaza Amjad, Mian Ashfaq Ali, Faisal Siddiqui, Aamir Mubashar, Muhammad Ahsan (2025)

Results in Engineering

PREDICTING SUSPECTED RADIATION NECROSIS IN BRAIN CANCER PATIENTS WITH A MACHINE LEARNING MODEL BASED ON THE PROTON LINEAL ENERGY SPECTRUM

Mark Newpower, Joseph Decunha, Radhe Mohan, Erik Traneus, Ozer Algan, Jerry Jaboin (2025)

International Journal of Particle Therapy

First Experimental Demonstration of Machine Learning-Based Tuning on the PSI Injector 2 Cyclotron

M. Haj Tahar, W. Joho, E. Solodko, M. Bocchio, S. Marquie, M. Busch, A. Barchetti, J. Grillenberger, J. Snuverink, M. Schneider (2025)

arXiv

Machine Learning-Based Prediction of Mechanical Properties of Recycled Aggregate Concrete

Dr. Chethan L S, Akash H E, Sanketh M B, Chinmay k karatekar, Revanth Kumar (2025)
International Journal of Scientific Research in Engineering and Management

First Experimental Demonstration of Machine Learning-Based Tuning on the PSI Injector 2 Cyclotron

Haj Tahar, M., Joho, W., Solodko, E., Bocchio, M., Marquie, S., Busch, M., Barchetti, A., Grillenberger, J., Snuverink, J., Schneider, M. (2025)

arXiv

First Experimental Demonstration of Machine Learning-Based Tuning on the PSI Injector 2 Cyclotron

Sandrine Marquié (2025)

arXiv (Cornell University)

Xopt and Badger: a machine learning ecosystem for real-time accelerator control and optimization

Roussel, Ryan, Kennedy, Dylan, Edelen, Auralee, Kuklev, Nikita, Miskovich, Sara, Zhang Zhe (2025)
JACOW

Long short-term memory of recurrent neural network to the analysis of temperature rise on the production target in Hadron Experimental Facility of J-PARC

Agari Keizo, Sato Yoshinori, Toyoda Akihisa, Morino Yuhei, Akiyama Hironobu (2025)
JACOW

Health state assessment of air compressor based on stacking ensemble learning

Dali Hou, Zhengrong Yang, Xiaoran Wang (2025)

Journal of Mechanical Science and Technology

Particle Tracking in the NA62 Experiment

Plini, Leonardo, Tinti, Gemma, Spadaro, Tommaso, Spinelli, Indro, Galasso, Fabio (2025)
PoS

Reconstruction of Neutron Spectra Using Silicon Carbide Detectors in Monoenergetic Fields with Machine Learning Approach

Enrica Belfiore, Achment Chalil, Mehdi Ben Mosbah, Rodolphe Antoni, Olivier Llido, Abdallah Lyoussi, Christophe Destouches, Tomáš Slavíček, Andrea Šagátová, Jean-Emmanuel Groetz (2025)

Springer Link (Chiba Institute of Technology)

Signal increase ratio prediction with CNNs

Marius Zeinhofer, Kent-Andre Mardal (2025)

Simula SpringerBriefs on Computing

Revolutionary Engineers: Learning, Politics, and Activism at Aryamehr University of Technology. Sepehr Vakil, Mahdi Ganjavi, and Mina Khanlarzadeh (Cambridge, MA: MIT Press, 2025). Pp. 236. \$50 paper. ISBN 9780262552196

Timothy Nunan (2025)

International Journal of Middle East Studies

Photonic Quantum-Accelerated Machine Learning

Rambach, Markus, Roy Abhishek, Gilchrist, Alexei, Sakurai, Akitada, Munro, William J., Nemoto Kae, White, Andrew G. (2025)

arXiv (Cornell University)

Photonic Quantum-Accelerated Machine Learning

Rambach, Markus, Roy Abhishek, Gilchrist, Alexei, Sakurai, Akitada, Munro, William J., Nemoto Kae, White, Andrew G. (2025)

arXiv (Cornell University)

Autonomous Pressure Control in MuVacAS via Deep Reinforcement Learning and Deep Learning Surrogate Models

Guillermo Rodriguez-Llorente, Galo Gallardo, Rodrigo Morant Navascués, Nikita Khvatkin

Petrovsky, Anderson Sabogal, Roberto Gómez-Espinosa Martín (2025)

arXiv

Machine learning-based prediction of magnet errors in storage ring light sources

Jianhao Xu (2025)

arXiv

ColliderML: The First Release of an OpenDataDetector High-Luminosity Physics Benchmark Dataset

Doña Elitez, Paul Gessinger, Daniel Murnane, Marcus Selchou Raaholt, Andreas Salzburger, Stine Kofoed Skov, Andreas Stefl, Anna Zaborowska (2025)

arXiv

Machine learning-based prediction of magnet errors in storage ring light sources

Xu, Jianhao (2025)

arXiv

Autonomous Pressure Control in MuVacAS via Deep Reinforcement Learning and Deep Learning Surrogate Models

Rodriguez-Llorente, Guillermo, Gallardo, Galo, Navascués, Rodrigo Morant, Petrovsky, Nikita Khvatkin, Sabogal, Anderson, Martín, Roberto Gómez-Espinosa (2025)

Journal

bb-Magazin international. Frames of Learning and Integration

Ali Osman (2025)

Berufsbildung

Enhancing Learning Outcomes in an Undergraduate Course of Smart Construction Major Through the Application of a Simulation Tool

Linlin Zhao, Zhansheng Liu, XueFeng Zhao, Yan Bao (2025)

Computer Applications in Engineering Education

Allaei (2026)
Materials Today Physics

Guided waves with machine learning for structural health monitoring: transparent features and Monte Carlo confidence

Brazalez Juan, Nabarrete Airton (2026)
Fracture and Structural Integrity

Machine Learning-Based Prediction of Weld Pool Features in Laser-Arc Hybrid Welding of Ultra-High-Strength Wear-Resistant Steels

Bin Zhang, Zhe Gao, Sitong Li, Haijun Pan, Zhenguang Liu, Zigang Chen, Hongyan Liu, Xinjun Shen, Zhenxing Li, Xiaonan Wang (2026)
Journal of Materials Engineering and Performance

Machine learning for quantum control and information processing in trapped-ion systems

LI JiaChong, DING GeYi, YUAN Quan, YUAN MingZhe, FENG Mang (2026)
SCIENTIA SINICA Physica, Mechanica & Astronomica

Machine Learning Based Efficiency Calculator (MaLBEC) for Nuclear Fusion Diagnostics

Kimberley Lennon, Chantal Shand, Gemma Wilson, Robin Smith (2026)
Journal of Fusion Energy

Design and Development of Perovskite Materials Using Machine Learning Techniques

Md. Motinur Rahman, Subrata Saha, Md. Ziaul Hasan Majumder, Mohammad Abu Sayid Haque, M. Khalid Hossain (2026)
Handbook of Perovskites

Development and Validation of a Machine Learning-Based Radiomics Model Using Ultrasound Image Features for Prostate Cancer Risk Stratification

Anli Zhao, Shunlan Du, Yanhong Du, Maoliang Zhang, Jiangfeng Wu, Zhengping Wang, Liyan Hu (2026)
Annali Italiani di Chirurgia

Evaluation of the accuracy of detecting C-shaped canals in mandibular second molars identified by cone-beam computed tomography on panoramic radiographs using artificial intelligence algorithms developed with deep learning methods

O. Uysal, M. Polat, H. M. Akgül (2026)
Oral Radiology

Explainable physics-based constraints on reinforcement learning for accelerator optimization

Jonathan Colen, Malachi Schram, Kishansingh Rajput, Armen Kasparian (2026)
Machine Learning: Science and Technology

Evaluating machine learned nuclear data precision in full core nuclear reactor Monte Carlo neutronics and computational efficiency analyses

Alexander Hashemi, Rafael Macián-Juan, Martin Ohlerich (2026)
Scientific Reports

A Machine Learning Framework for ICU and Medical-Surgical COVID-19 Admission Forecasting

Kevin Geng, Ishaan Gupta, Sai Bharadwaj, Dylan Lam, Atharv Rao, Rajveer Grover, Dhruva Kanna, Devansh Karavati, Akshainie Pandella (2026)
IJARCCE

Comparative evaluation of machine learning models for thrust force estimation in aluminum 5083 H111

Emre Teke, Neslihan Özsoy, Deniz Demircioğlu Diren, Murat Özsoy (2026)
Materials Testing

Application of MULTIMEDIA Powerpoint Assisted Learning to Improve Learning Outcomes of Knowledge of the function of Fracture Machine Parts

Prastiyo Pras, Muhammad Khumaedi, Kriswanto (2026)

Jurnal Pendidikan Teknik Mesin

Sintering Process Analysis of Aluminum Matrix Composites Using Machine Learning

Kenjiro Sugio, Yuuki Shinohara, Yoshikazu Hayashi, Gen Sasaki (2026)

Defect and Diffusion Forum

Research on the application of machine learning algorithms in aeromagnetic survey

Li Liu, Lei Zhou, Qingfeng Xu (2026)

Seventh International Conference on Geoscience and Remote Sensing Mapping (GRSM 2025)

Identification of Lightning Strikes to Towers Using Their Electric Field Signatures and a Machine Learning Approach

Ziqin Ding, Si Chen, Hanbo Yang, Kang Yang, Vladimir A. Rakov, Joel B. Harley, Yanan Zhu, Istvan Kereszy (2026)

IEEE Sensors Journal

Robust Leakage Localization in Radioactive Material Transport via Multitask Learning

Yafei Chen, Shaohua Hu, Kaijing Yang, Hualuo Sheng, Chaofeng Wang, Yaping Wan (2026)

IEEE Sensors Journal

TTBO-FL: Joint Training, Trajectory, and Beamforming Optimization for Energy-Efficient Federated Learning in UAV Swarm

Chunyu Zhang, Yanlu Li, Yiming Liu, Yuzhen Huang, Zhi Zhang (2026)

IEEE Internet of Things Journal

Radar Signal Classification with Quantum Machine Learning: Ansatz Depth Impact on Expressibility

Gabriel F. Martinez, Alberto Croci, Francesco Drago, Alessandro Niccolai, Marco Mussetta, Riccardo E. Zich (2026)

Electronics

Technology Topic Mining and Patent Data Analysis of NdFeB Permanent Magnet Materials Based on Machine Learning

Jian-zhong LIU, Yang BAI, Xin-fang ZHANG, Xiao-yu LIU, Zhen-jun JIN, Ke WANG, Meng-cheng ZHOU (2026)

Chinese Rare Earths

Predicting fire resistance of reinforced concrete (RC) and concrete-filled steel tubular (CFST) columns through machine learning

Muhammad Noman, Muhammad Salman, Mati Ullah, Muhammad Faizan, Afaq Ahmed (2026)

Innovative Infrastructure Solutions

Support Vector Machines

Yinglin Xia, Jun Sun (2026)

Machine Learning for Microbiome Statistics

U-Net-based surrogate modeling for attosecond X-ray free-electron lasers

Yufei Wei, Bingyang Yan, Chenzhi Xu, Jiawei Yan, Haixiao Deng (2026)

arXiv

Using Machine Learning to Predict First-Order Reaction Rate Constants of PFAS Degradation

Chenhao Pei, Yifan Qian, Jie Shen, Naichi Zhang, Qi Liu, Zhiqiang Li, Shuailei Pu, Tongliang Wu, Yujun Wang, Cun Liu (2026)

Bulletin of Environmental Contamination and Toxicology

Thermal Runaway of Chemical Reactors: An Experimental, Modeling, and Machine-Learning Investigation

Tobias K. Misicko, Joaquin Herrero, Jacob C. Robinson, Daniela S. Mainardi, Yang Xiao (2026)

Industrial & Engineering Chemistry Research

Towards continual machine learning for particle accelerators

Rajput, Kishansingh, Schram, Malachi, Blokland, Willem, Zhukov, Alexander, Lin, Sen (2026)
JACoW

Integrating physics and machine learning for unified seismic forward modeling and reservoir property inversion

Yeheya Zayier, Yaxiaer Yalikun, Yuanfeng Cheng, Dinghui Yang (2026)
Scientific Reports

Semi-supervised learning for dose prediction in targeted radionuclide therapy: a synthetic data study

Jing Zhang, Alexandre Bousse, Chi-Hieu Pham, Kuangyu Shi, Julien Bert (2026)
Physics in Medicine & Biology

Vibrational characterization and machine learning-based classification of defects in TWCNTs

Jalpa Ardeshana, Bhavik Ardeshana, Umang Jani, Jay Vala, Ajay Patel, Hitesh Shah, Vimal Vaghela (2026)
Mechanics Based Design of Structures and Machines

ReliaLearnR: Learning Modules for Reliability Analysis

Paul B. Govan (2026)
Journal of Open Source Education

Advancing shear behavior prediction in SFRC beams: a comparative study of machine learning techniques

Esayies A. Wassie, Leule M. Hailemariam, Habtamu A. Tadesse, Temesgen A. Kassie (2026)
Discover Civil Engineering

A machine learning based approach to online electron reconstruction at CLAS12

Tyson, R., Gavalian, G. (2026)
Comput.Phys.Commun.

Reinforcement learning for accelerator beamline control: A simulation-based approach

Anwar Ibrahim, A. Petrenko, Maxim Kaledin, Ehab S Suleiman, F. Ratnikov, ■. ■■■■■■■■ (2026)
International Journal of Modern Physics E

Machine Learning Radiomics in Computed Tomography for Prediction of Tumor and Nodal Stages in Colorectal Cancer

Lara de Souza Moreno, Tony Alexandre Medeiros da Silva, Mayra Veloso Ayrimoraes Soares, João Luiz Azevedo de Carvalho, Fabio Pittella-Silva (2026)
Cancers

Attention-Driven Feature Extraction for XAI in Histopathology Leveraging a Hybrid Xception Architecture for Multi-Cancer Diagnosis

Shirin Shila, Md. Safayat Hossain, Md Fuyad Al Masud, Mohammad Badrul Alam Miah, Afrig Aminuddin, Zia Muhammad (2026)
Machine Learning and Knowledge Extraction

Automatic detection of loose bolts in pipeline structures based on anti-noise mel cepstrum differential features and deep learning

Yixuan Chen, Hongyi Chen, Jingyi Wei, Jianchao Wu (2026)
Journal of Civil Structural Health Monitoring

JuTrack, a Julia-based auto-differentiable accelerator simulation code for advanced dynamics, scientific machine learning and optimization

Wan, Jinyu, Alamprese, Helena, Ratcliff, Christian, Qiang, Ji, Hao, Yue (2026)
JACoW

Data-driven beam diagnostics using betatron radiation

Saberi, Hossein, Burvill, Jasper, Siddle, Carys, Xia, Guoxing, Patrick Farmer, John, Pukhov, Alexander (2026)
Plasma Phys.Control.Fusion

Seismic Response Estimation and Performance Assessment of Large-Scale Structures Using Deep-Learning Methods

Jacob A. Morgan, Kalil Erazo, Oral Buyukozturk (2026)
Journal of Structural Engineering

Toward closed-loop quality assurance in powder bed fusion additive manufacturing: Defect detection, machine learning, and computational modeling

Mohammad Taghian, Ali Pilehvar Meibody, Abdollah Saboori, Luca Iuliano (2026)
Journal of Manufacturing Processes

PRISM: Distribution-free Adaptive Computation of Matrix Functions for Accelerating Neural Network Training

Shenghao Yang, Zhichao Wang, Oleg Balabanov, N. Benjamin Erichson, Michael W. Mahoney (2026)

arXiv (Cornell University)

PRISM: Distribution-free Adaptive Computation of Matrix Functions for Accelerating Neural Network Training

Shenghao Yang, Zhichao Wang, Oleg Balabanov, N. Benjamin Erichson, Michael W. Mahoney (2026)

arXiv (Cornell University)

Optimization of the Claus Process Using Machine Learning and Real Operational Data: A Case Study in an SRU

Amirfarzam Shokouhalaei (2026)
Chemistry Africa

Axial-stress classification of steel pipes from magnetic remanence via topology-aware multi-probe graph learning: controlled full-scale validation

He Dang, Laibin Zhang, Jianchun Fan, Fumin Gao, Rui Li, Huadong Song, Wenguang Hu (2026)
Nondestructive Testing and Evaluation

High-Resolution Image Reconstruction with Unsupervised Learning and Noisy Data Applied to Ion-Beam Dynamics for Particle Accelerators

Francis Osswald, Mohammed Chahbaoui, X. Liang (2026)
Journal of Instrumentation

A Hybrid Monte Carlo–Machine Learning Framework for High-Energy Neutron Shielding Using Boron-Enhanced Concrete

Demet Sariyer, Elif Yildirim (2026)
International Journal of Computational and Experimental Science and Engineering

Current applications of artificial intelligence and machine learning in oncologic ICU management

Shivam Dubey (2026)
Onco Critical Care

The Research on Chain Fault Prediction in AC-DC Hybrid Power Grids Based on Deep Learning

Wenchao Qin, Xufei Liu, Yiran Tao (2026)
Springer Science and Business Media LLC

A Study on Predicting Engine Performance Outputs by Machine Learning Algorithms in a Single Cylinder HCCI Engine

Ahmet Çelik, Mehmet Akif Kunt (2026)
Engineering Perspective

Advanced quality assessment and damage monitoring in 3D-printed plastic and composite beams using machine learning

Gia Hoang Phan (2026)
Progress in Additive Manufacturing

Hybrid machine learning-monte carlo approach for neutron shielding in high-energy proton facilities

Sariyer, Demet (2026)
J.Subatomic Part.Cosmol.

Advanced deep learning techniques for classifying dental conditions using panoramic X-ray images

Alireza Golkarieh, Bahareh Afjehsoleymani, Kiana Kiashemshaki, Sajjad Rezvani Boroujeni (2026)
BMC Oral Health

Multimodal Learning for Arcing Detection in Pantograph-Catenary Systems

Hao Dong, Eleni Chatzi, Olga Fink (2026)
arXiv

Deep-learning-based Ultrasonic Guided Wave Detection of Turnout Switch Rail Cracks

ChengYue Lv, Ping Wang, Hao Sun (2026)
Sensors and Materials

Digital twin framework for PIP-II linac: AI-driven multi-scale modeling from ion source to 800 MeV

Pathak, Abhishek, Hanlet, Pierrick, Miceli, Tia (2026)
JACoW

Machine learning-enhanced infrared imaging for temperature anomaly detection in power supplies

Mohsen, Osama, Lobach, Ihar, Borland, Michael, Sun, Yine (2026)
JACoW

Anomaly detection of slow-moving variables at LANSCE for improved beam quality

Huang, En-Chuan, Yampolsky, Nikolai (2026)
JACoW

Online optimizations of NSLS-II Linac and Linac-to-Booster beam lines using machine learning methods

Song, Minghao, Wang, Guimei, Hidaka, Yoshiteru, Yang, Xi (2026)
JACoW

Machine learning-driven computations of 3D Coherent Synchrotron Radiation

Leon, Christopher, Scheinker, Alexander, Yampolsky, Nikolai, Anisimov, Petr M. (2026)
JACoW

Safe extremum seeking for real-time adaptive accelerator control

Williams, Alan, Scheinker, Alexander (2026)
JACoW

Machine learning assisted Bayesian calibration of accelerator digital twin from orbit response data

Lin, Weijian, Kelly, Christopher, Hoffstaetter, Georg, Brown, Kevin, Urban, Nathan (2026)
JACoW

Integrating simulation and machine learning for Proton Storage Ring beam analysis

Leon, Christopher, Huang, En-Chuan, Scheinker, Alexander, Anisimov, Petr M. (2026)
JACoW

Towards differentiable beam dynamics modeling in BLAST/ImpactX

Huebl, Axel, Mitchell, Chad, Lehe, Remi, Charleux, Grégoire, Myers, Andrew, Zhang, Weiqun, Qiang, Ji, Vay, Jean-Luc, Kaiser, Jan, Hespe, Christian, Gonzalez-Aguilera, Juan Pablo, Xu, Chenran, Santamaria Garcia, Andrea, Roussel, Ryan, Edelen, Auralee, Moses, William Steven (2026)
JACoW

A self-supervised transformer for RF cavity signal denoising

Rajesh, Vikshar, Edelen, Auralee, Hall, Christopher, O'Shea, Finn, Edelen, Jonathan (2026)

JACoW

Machine learning using beam loss monitors for DIAMOND-II

Lehmann, Corey, Bobb, Lorraine, Burrows, Philip (2026)

JACoW

Beam diagnostics, data acquisition system, and applications of machine learning at the KEK e-/e+ Linac

Miyahara, Fusashi, Kinoshita, Kaito, Kurata, Masakazu, Iwasaki, Masako, Natsui, Takuya, Okayasu, Yuichi, Satake, Itsuka, Satoh, Masanori, Uemura, Kosuke, Wang, Di (2026)

JACoW

PRISM project's update: making spatial machine learning more reliable, plus related activities

Jakub Nowosad (2026)

Front Matter

Ultraefficient Compressed Phonocardiogram Classification on a Custom Embedded Neural Accelerator

Domenico Ragusa, Rens Baeyens, Danilo Pau, Elisa Marenzi, Jan Steckel, Walter Daems, Francesco Leporati, Emanuele Torti (2026)

IEEE Internet of Things Journal

CGMAXPRA19-CS3-1-ENG PDF Dumps – Trusted Learning Resource

Blade Shark (2026)

ResearchHub Technologies, Inc.

Machine Learning for Anomaly Detection in Neural Network Security and SRF Cavities

Hal Ferguson (2026)

ODU Digital Commons (Old Dominion University)

Continual Learning for Particle Accelerators

Nuclear Physics (NP) USDOE Office of Science (SC), M. Schram, Thomas Jefferson National Accelerator Facility (TJNAF), Sen Lin, Kishansingh Rajput, Alexander Zhukov, Willem Blokland (2026)

Unknown Venue

Architectural Taxonomy of AI Accelerators: A Memory and Interconnect Centric Perspective

Nour Abu Jassar (2026)

Unknown Venue

Real-world evaluation of automated defect detection in masonry bridges using 360° imagery with machine learning

Arijit Sen, Quanbin Sun, Song Wu, Saeed Talebi (2026)

Construction Innovation

Integrating Industry 5.0 principles into electrical machine design

Pankaj Kumar Keshri (2026)

Artificial Intelligence-Powered Electrical Machine Design

Türkiye Yüzyıllı Maarif Modeli Okul Öncesi Eğitim Programı Öğrenme Çıktıları ile 2024 Erken Çocukluk Eğitimi Programının Kazanım ve Göstergelerinin 21. Yüzyıl Becerilerine Yer Verme Durumu Açısından Karşılaştırılması

Gülden ERDOĞAN, Vedat BAYRAKTAR (2026)

Anadolu Kültürel Araştırmalar Dergisi

Development and implementation of the IMRA multiplatform Foundations in Robotic Surgery online learning curriculum

Tayla Fay, Daniel Costello, Dean Driscoll, Niall M. Corcoran, Anthony J. Costello, Henry Woo, Helen Mohan (2026)

Journal of Robotic Surgery

Diagnosing Induction Motor Fault Based on ReliefF Feature Selection Algorithm and Support Vector Machine Model

Felly Anta, Didik Djoko Susilo, Zainal Arifin (2026)

Applied Mechanics and Materials

Advancing accelerator virtual beam diagnostics through latent evolution modeling: an integrated solution to forward, inverse, tuning, and UQ problems

Rautela, Mahindra, Scheinker, Alexander (2026)

arXiv

Advancing accelerator virtual beam diagnostics through latent evolution modeling: an integrated solution to forward, inverse, tuning, and UQ problems

Mahindra Rautela, Alexander Scheinker (2026)

arXiv

Machine learning at the Spallation Neutron Source accelerator and target

Armen Kasparian, Willem Blokland, Alexander Zhukov, Anant Ray, Drew Winder, Kishansingh Rajput, Malachi Schram (2026)

JACOW

Machine learning-enhanced deterministic controls in lasers and accelerators

Dan Wang, Alessio Amodio, Alex Picksley, A. J. Gonsalves, Curtis J. Berger, Hai-En Tsai, Jeroen van Tilborg, Mahek Logantha, N. R. Vora, Qiang Du, Russell Wilcox, Samuel K. Barber, Tong Zhou, Z. Eisentraut (2026)

JACOW

Educational Effects of an AI Mobility Accelerator Platform (aMAP)-Based Autonomous Driving Competition for High School Students

Chan Seob Park, Kuk Won Ko (2026)

Asia-pacific Journal of Convergent Research Interchange

MECD+: Unlocking Event-Level Causal Graph Discovery for Video Reasoning

Tieyuan Chen, Huabin Liu, Yi Wang, Yihang Chen, Tianyao He, Chaofan Gan, Huanyu He, Weiyao Lin (2026)

IEEE Transactions on Pattern Analysis and Machine Intelligence

Temporal-frequency enhanced fourier neural operator and collaborative adversarial learning for multi-physics-coupled transient simulation of natural gas pipelines

Yuqian Wang, Gang Liu, Xiaojing Zhu, Jiaying Zou, Lei Chen (2026)

Expert Systems with Applications

Optimizing pulsed field ablation for cardiac arrhythmias integrating Taguchi method, machine learning and genetic algorithms

Ethan Nabuurs, Kuljeet Singh Grewal, Amara Sanchez, Mikkayla Ellsworth-Reid, Robert F. Gilmour, Roderick Melnik, Sundeep Singh (2026)

Results in Engineering

X-ray CT analysis of internal defects in 3D-printed continuous carbon fiber composites using deep learning

Runze Yang, Jinyue Zhang (2026)

Materials Letters

High-energy neutron shielding analysis using curve fitting and machine learning: a Monte Carlo-based study in concrete and Ferro-Boron configurations

Ali Osman GÖKCAN, Demet Sarıyer (2026)

Annals of Nuclear Energy

PAC Guarantees for Reinforcement Learning: Sample Complexity, Coverage, and Structure

Joshua Steier (2026)

arXiv

Analisis Sentimen Kontaminasi Radioaktif di Kawasan Industri Cikande Menggunakan Algoritma Support Vector Machine

Taufik Ramlan Alfiansyah, Audy Abdillah Hidayat, Alfarezi Hidayat Pratama, Agil Aqshol Mahenda, Muhammad Rafly (2026)
Jurnal Informatika Terpadu

Support Vector Machine and k-Means Clustering for Advanced Wheel Flat Identification: A Comparison of Supervised and Unsupervised Methods

Alireza Chegini, Mohammadreza Mohammadi, Araliya Mosleh, Cecilia Vale, Ramin Ghiasi, Ruben Silva, Antonio Guedes, Andreia Meixedo, Abdollah Malekjafarian (2026)
Machines

Hybrid analytical-machine learning model for data-efficient prediction of ultrafast laser ablation

Eric P. Mottay (2026)
Laser-based Micro- and Nanoprocessing XX

Enhancing the extrapolative capabilities of deep learning models on temporal parameter sets in femtosecond laser processing

Shuntaro Tani (2026)
Laser-based Micro- and Nanoprocessing XX

Solid-state laser design process empowered by machine learning and deep neural networks

John Vetrovec, Carey A. Smith, David M. Filgas, Drew A. Copeland, Cassidy S. Vetrovec (2026)
Solid State Lasers XXXV: Technology and Devices

Machine learning-based modeling for structural modification induced by femtosecond laser pulses

Shuntaro Tani (2026)
Nanoscale and Quantum Materials: From Synthesis and Laser Processing to Applications 2026

“War Machine” 2026 Srteaming-Ita CB01 [Altadefinizione]

coeth mitch (2026)
ResearchHub Technologies, Inc.

[Altadefinizione] “War Machine” 2026 Streaming-Ita CB01

coeth mitch (2026)
ResearchHub Technologies, Inc.

NOBLE: Accelerating Transformers with Nonlinear Low-Rank Branches

Ethan Smith (2026)
arXiv

High-Resolution Image Reconstruction with Unsupervised Learning and Noisy Data Applied to Ion-Beam Dynamics for Particle Accelerators

Francis Osswald, Mohammed Chahbaoui, Xinyi Liang (2026)
arXiv

GNN For Muon Particle Momentum estimation

Vishak K Bhat, Eric A. F. Reinhardt, Sergei Gleyzer (2026)
arXiv

Characterization and upgrade of a quantum graph neural network for charged particle tracking

Matteo Argenton, Laura Cappelli, Concezio Bozzi (2026)
arXiv

Method for reconstruction the energy spectrum of a linear electron accelerator using neural networks

A.A. Kim, F.R. Studenikin, V.V. Khankin, S.A. Zolotov, A.P. Lyubomudrov, A.P. Chernyaev (2026)
Radiation Physics and Chemistry

5 Künstliche Intelligenz, Machine Learning und Deep Learning in attributiven Prüfprozessen (2026)

Attributive Prüfprozesse in der Produktion

A Physics-informed Deep Learning Model for the Magnetic Classification of Solar Active Regions

Ming Li, Siwei Liu, Jingjing Wang, Yanmei Cui, Bingxian Luo, Xin Wang (2026)
The Astrophysical Journal Supplement Series

Hardware Architectures and Optimization Techniques for Convolutional Neural Network Accelerators

Hemkant Nehete, Gaurav Verma, Amit Monga, Alok Kumar Shukla, Shailendra Yadav, Brajesh Kumar Kaushik (2026)
Unknown Venue

Learning post-Newtonian corrections from numerical relativity

Anonymous (2026)
Physical Review D

ANHP: Adaptive Neural Hawkes Processes for Causal Structure Learning on Event Sequences

Yongjian Chang, Duxin Chen, Yujin Cai, Wenwu Yu (2026)
IEEE Internet of Things Journal

Machine learning-based prediction of the structural performance of a long-bolted steel connection

Mostafa Elhadary, Ahmed Bediwy, Ahmed Elshaer (2026)
AI in Civil Engineering

Noise reduction for self-emission x-ray images in ICF using a physics-informed deep learning framework

Rui Hua, Tianzhu Gong, Chuankui Sun, Chengmin Lan, Yanxu Su, Liang Guo, Changyin Sun (2026)
Nuclear Fusion

Verification and Validation of Physics-Informed Surrogate Component Models for Dynamic Power-System Simulation

Petros Ellinas, Indrajit Chaudhuri, Johanna Vorwerk, Spyros Chatzivasileiadis (2026)
arXiv

Comprehensive Structural Descriptors Enable Machine Learning Prediction of Perovskite Band Characteristics

Zixuan Ni, Zhirui Zhang, Yiming Chen, Daye Zheng, Jian Xu (2026)
The Journal of Physical Chemistry Letters

Learning curves and training thresholds for paediatricians performing neonatal hip ultrasound using the Graf method in a universal screening program

I. Miras Aguilar, L. Fidalgo Marrón, M. Pangua Gómez, C. Gilarte Herrera, C. Olteanu Olteanu, E. Castaño Andreu, C. Llorente Ruiz, A. Andrés Bartolomé, A. Ortigado Matamala (2026)
European Journal of Pediatrics

AI-Ready Control System for the Fermilab Accelerator Complex

Tia Miceli, Erik Gottschalk, Donovan Tooke, Evan Milton, Robert Santucci, Hayden Hoschouer, Michael Balcewicz, Jennifer Case, Abhishek Deshpande, Kit Fieldhouse, Sudeshna Ganguly, Beau Harrison, Aisha Ibrahim, Thomas Kobilarcik, Michael Olander, Abhishek Pathak, Jason St. John, Aaron Sauers (2026)
arXiv

Efficient Two-Stage Sequential Machine Learning Model for Progressive Damage Analysis in Fiber-Reinforced Composites

Mohammadamin Alibakhshi, Parisa Khodabakhshi (2026)
Machine Learning: Engineering

Prediction of phase diagrams using machine learning backed nested sampling

Nico Unglert (2026)
repositUM (TU Wien)

Mesh-Free Modeling of Heat Transfer Dynamics for Rapid Assessment of Necrosis Zones in Hepatic Tumor Radiofrequency Ablation Using Physics-Informed Neural Networks

Muhammet Kaan Yeşilyurt (2026)
Acadlore Transactions on AI and Machine Learning

Reinforcement Learning: A Comprehensive Open-Source Course

Ali Hassan Ali Abdelwanis, Barnabas Haucke-Korber, Darius Jakobeit, Wilhelm Kirchgässner, Marvin Meyer, Maximilian Schenke, Hendrik Vater, Oliver Wallscheid, Daniel Weber (2026)
Journal of Open Source Education

Fast 3D whole-body occupational dose estimation in interventional radiology using physics-informed deep learning

Hussein Harb, Didier Benoit, Chi-Hieu Pham, Bahaa Nasr, Julien Bert (2026)
Radiological Physics and Technology

Machine learning-based virtual diagnostics of dielectric laser acceleration

Thilo Egenolf, Oliver Boine-Frankenheim (2026)
arXiv

Machine learning-based virtual diagnostics of dielectric laser acceleration

Thilo Egenolf, Oliver Boine-Frankenheim (2026)
arXiv (Cornell University)

Data-Driven Optimisation of Superconducting Magnets at CEA Paris-Saclay

Damien F. G. Minenna, Guillaume Dilasser, Robin Penavaire, Valerio Calvelli, Thibault de Chabannes, Thibault Leclerc, Thomas Achard, Jason Le Coz, Christophe Berriaud, Benoît Bolzon, Antomne Caunes, Phillipe Fazilleau, Hélène Felice, Clément Genot, Antoine Guinet, Nikola Jerance, François-Paul Juster, Thibaut Lemercier, Gilles Lenoir, Clément Lorin, Yann Perron, Camille Pucheu-Plante, Étienne Rochepault, Damien Simon, Francesco Stacchi, Michel Segreti, Vincent Trauchessec, Olivier Tuske, Hajar Zgour (2026)
arXiv

Reduced dataset-based machine learning model for blast damage assessment of reinforced concrete columns

Yeeun Kim, Kihak Lee, Jiuk Shin (2026)
Engineering Structures

Machine learning-based virtual diagnostics of dielectric laser acceleration

Thilo Egenolf, Oliver Boine-Frankenheim (2026)
arXiv (Cornell University)

Multiple-Class Classification of a Diesel Locomotive Engine's Mechanical Condition Based on Wavelet Properties of Vibration Signals via Machine-Learning Techniques

Andrey Belyaev (2026)
Bulletin of scientific research results

Microsecond-latency feedback at a particle accelerator by online reinforcement learning on hardware

L. Comparin, Michele Caselle, Andrea Santamaria Garcia, Chenran Xu, Edmund Blomley, Timo Ditschler, Akira Mochihashi, Marcel Schuh, Johannes Leonhard Steinmann, Erik Bründermann, Andreas Kopmann, Jürgen Becker, A.-S. Mueller, Marc Weber (2026)
Machine Learning Science and Technology

Publisher Correction: Fine-Tuning Universal Machine-Learned Interatomic Potentials for Applications in the Science of Steels

Naveen K. Mohandas, Sebastián Echeverri Restrepo, Marcel H. F. Sluiter (2026)
Journal of Phase Equilibria and Diffusion

Machine-State Embeddings as an Operational Coordinate System for Accelerator Operation

Chris Tennant, Jundong Li, Song Wang (2026)

arXiv

Deep learning-based high-speed railway communication systems

Do Viet Ha, Trinh Van Chien, Hien Quoc Ngo (2026)

Scientific Reports

Impact of Problem-Based Learning On Interest and Achievement in Mechanics Among Senior Secondary Physics Students in Fct, Abuja

(2026)

Iconic Research and Engineering Journals

Adaptive Control Meets Deep Learning: EXTREMUM SEEKING IN LATENT SPACE WITH PARTICLE ACCELERATOR APPLICATIONS

Alexander Scheinker (2026)

IEEE Control Systems

Machine learning approach to estimation of in vivo measured dose and treatment planning for total body irradiation

Sangmin Lee, Jung-in Kim, Seonghee Kang, Jaeman Son, Hyeongmin Jin, Chang Heon Choi, Jong Min Park, Joo Ho Lee, Ji Hyun Chang, Hyun-Cheol Kang, Seongmoon Jung (2026)

Radiation Oncology

Deep learning for carotid Doppler spectra classification

Charita Bhikha, Kahesh Dhuness, Mathilda Mennen, Normé Jamieson-Luff, Ntobeko A. B. Ntusi, Richard Wheatley (2026)

Medical & Biological Engineering & Computing

SkillSync: An AI-Based Skill Gap Analyzer Using Machine Learning

ARTHI. K, MAHALAKSHMI .B, THAMIZHSELVI. J (2026)

INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING AND MANAGEMENT

A Coupling Contrastive Learning with Biggest Margin Trees for Classifying Chest X-ray Images

Thanh-Nghi Do, Tri-Thuc Vo (2026)

Vietnam Journal of Computer Science

A PSO-random forest-based cross-domain fault pattern recognition system for high-speed train bearings

Junjie Lin, Zhuofu Deng (2026)

International Conference on Machine Vision, Intelligent Imaging, and Pattern Recognition (MVIIPR 2025)

Ultrasound■ MRI Fusion Deep Learning Radiomics Nomogram: Refining Prognostic Stratification for Triple■Negative Breast Cancer Treated With Neoadjuvant Chemotherapy

Yang Liu, Wei Wu (2026)

Journal of Ultrasound in Medicine

Assessment of machine learning models for predicting the deflection of concrete beams

Arvindan Sivasuriyan, Dhanasingh Sivalinga Vijayan, Roman Trach, Hydayatullah Bayat, Artur Góral, Anna Pi■tocha, Eugeniusz Koda (2026)

Scientific Reports

A physics-guided temporal convolutional learning with XGBoost integration for degradation-aware and robust electric vehicle range prediction

Kaliappan Esakkiappan, P. Gowtham, V. Beula Shanthi Ammani Ammal, D. Shalini, T. Mariprasath, Seif Al Bustanji, Viktoriia Berezhnychenko (2026)

Scientific Reports

Research on mechanical properties prediction of regenerative ultra-high performance Concrete based on machine learning 1

Prathmesh Shinde (2026)
International Journal for Research in Applied Science and Engineering Technology

Multi-sensor Fusion with Hybrid Deep Learning Convolutional Neural Network-Long Short Term Memory for Real-time Welding Activity Recognition and Occupational Safety Compliance

(2026)

International Journal of Intelligent Engineering and Systems

Development of an Experimental Test Stand and Machine Learning Surrogate Models for Capturing Complex Responses in Pulsed Septa

Kawa, Krzysztof, Szumlak, Tomasz, Yagci, Omer Yusuf, Balhan, Bruno, Lackner, Friedrich (2026)
IEEE Trans.Appl.Supercond.

Machine learning for mapping multipactor susceptibility in RF systems: Capabilities and generalization constraints

Asif Iqbal, John Verboncoeur, Peng Zhang (2026)

APL Machine Learning

How to Overcome Limitations of the Abaqus Superelasticity Material Model with Reduced Order Modeling and Machine Learning

M. Fassin, C. Degel, P. Hempel, K. Schulz (2026)

International Conference on Shape Memory and Superelastic Technologies

Machine learning time propagators for time-dependent density functional theory simulations

Karan Shah, Attila Cangi (2026)

Machine Learning: Science and Technology

Machine learning-driven inverse design of cosine beam honeycombs for zero Poisson's ratio and predetermined mechanical curves

Zi-Tao Wang, Hai-Tao Liu (2026)

International Journal of Applied Mechanics

Comparative efficacy of supervised and semi-supervised machine learning classifiers for energy-based discrimination of near-fault pulse-type ground motion

Sayed S. R. Moustafa, M. Sami Soliman, Shimaa H. Elkhoully, Mohamed Yassien (2026)

Earth Science Informatics

Defect prediction for magnetorheological damper using a deep learning-based CLSTM classifier

Yoon-Mok Kim, Jong-Seok Oh, Bo-Gyu Kim (2026)

Journal of Mechanical Science and Technology

Navigating learning analytics: challenges, opportunities, and recommendations

Marte Blikstad-Balas (2026)

Handbook of Learning Analytics and Adaptive Learning in Schools

Reinforcement Learning Concepts

Jonas Hellgren, Johannes Lindgren (2026)

Reinforcement Learning Explained

Leveraging reinforcement learning, genetic algorithms and transformers for background determination in particle physics

Guillermo Hijano Mendizabal, Davide Lancierini, Alex Marshall, Andrea Mauri, Patrick Haworth Owen, Mitesh Patel, Konstantinos Petridis, Shah Rukh Qasim, Nicola Serra, William Sutcliffe, Hanae Tilquin (2026)

Physica Scripta

Machine Learning-Based Graph Simplification for Symbolic Accelerators

Tiffany Yu, Rye Stahle-Smith, Darssan Eswaramoorthi, Rasha Karakchi (2026)

arXiv (Cornell University)

Integrated image processing and machine learning framework for defect detection and prediction in selective laser melting of Hastelloy X

M. Memari, A. Farzadi, H. Davarzani, A. Shamsipoor (2026)

Welding in the World

A machine-learning-based approach for railway excitation force prediction

Hauke Schmidt, Gang Chen, Guozhen Jing, Daniel Zinken (2026)

Railway Sciences

Machine Learning-Based Graph Simplification for Symbolic Accelerators

Tiffany Yu, Rye Stahle-Smith, Darssan Eswaramoorthi, Rasha Karakchi (2026)

ArXiv.org

Koopman-Operator-Based Control of Hypersonic Flight Vehicles With Few-Shot Learning for Internet of Aerospace Things

Wenjia Deng, Tingting Wang, Wencheng Zou, Jian Guo, Zhengrong Xiang (2026)

IEEE Internet of Things Journal

A Multi-Task Deep Learning Model for Quality Control of Periapical Radiographs: Simultaneous Technical Error Classification and Quality Grading

Wen Fang, Leqi Liu, Xiaokan Wang, Jun Wang, Yuqing Meng, Keying Wu, Tie Liu, Bin Feng (2026)

Dentomaxillofacial Radiology

Advanced Machine Learning Enabled Modern Power System

Muhammad Asim Amin (2026)

CINECA IRIS Institutional Research Information System (University of Genoa)

Quantum Phenomena

John N. Moyer (2026)

Creating Learning Systems for Higher Education

A Sharper Picture of Generalization in Transformers

Paul Lintilhac, Sair Shaikh (2026)

arXiv

The diffusion-driven orthorhombic to tetragonal transition in YBa₂Cu₃O₇ derived with a machine learning interatomic potential

Davide Gambino, Niccolò Di Eugenio, Jesper Byggmästar, Johan Klarbring, Daniele Torsello, Flyura Djurabekova, Francesco Laviano (2026)

npj Quantum Materials

Multi-class anatomical landmark detection in periapical radiographs with deep learning

Cansu Buyuk, Alperen Saruhan, Fatma Yuce, Ozer Celik, Elif Bilgir, ■brahim Sevki Bayrakdar (2026)

Odontology

A complete discussion on fully reconfigurable, digital, scalable, graph and sparsity-aware near-memory accelerator for graph neural networks

Siddhartha Raman Sundara Raman, LIZY JOHN, Jaydeep P. Kulkarni (2026)

arXiv (Cornell University)

A complete discussion on fully reconfigurable, digital, scalable, graph and sparsity-aware near-memory accelerator for graph neural networks

Siddhartha Raman Sundara Raman, LIZY JOHN, Jaydeep P. Kulkarni (2026)

ArXiv.org

Automated orbital wall fracture detection and severity classification of ocular injuries using deep learning

Farhad Salari, Zahra Meskar, Farhad Fallah, Maryam Fazeli, Reza Samiee, Hossein Arabalibeik, Seyed Mohsen Rafizadeh (2026)

Scientific Reports

Development of a deep-learning model for classification of intensivists' extubation decisions using ventilator graphic monitor images: a prospective observational study

Tatsuya Hayasaka, Masaki Nakane, Yuka Fujiwara, Yoko Yuzawa, Ryuto Yokoyama, Kiyotaka Soekawa, Masahiro Kuroki, Kenya Yarimizu, Yu Onodera, Tadanori Fukami, Hiroaki Toyama (2026)

Journal of Intensive Care

Advanced Machine Learning Enabled Modern Power System

Muhammad Asim Amin (2026)

CINECA IRIS Institutional Research Information System (University of Genoa)

Automatic scheduling of satellite tracking tasks by means of policy-gradient reinforcement learning and transformer-based pointer networks

Ravi Dobariya, Thomas Hobiger (2026)

CEAS Space Journal

Machine Learning methods for event classification and vertex reconstruction of the 12C + 12C reaction with the MATE-TPC

Minghui Zhang, Xiaobin Li, Jie Chen, Ningtao Zhang, Fenhua Lu, Junrui Ma, Jiazhen Yan, Wanqin Tu, Xiaodong Tang, Bingshui Gao, Chengui Lu, Zhichao Zhang, Jinlong Zhang, Weiping Liu (2026)
arXiv

Rapid Secondary Neutron Dose Estimation in Proton Accelerators Using FLUKA and Machine Learning

Demet Sariyer, Elif Yıldırım (2026)

Süleyman Demirel Üniversitesi Fen Edebiyat Fakültesi Fen Dergisi

Classification of PSQA outcomes in prostate VMAT treatments: a comparative study of machine learning models

Francis C. Djoumessi Zamo, Alexandre Ngwa Ebongue, Daniel Bongue, Maurice Moyo Ndontchueng, Christopher F. Njeh (2026)

Biomedical Engineering Advances

Lightweight CNN-Based Anomaly Detection for High Voltage Converter Modulators in the Spallation Neutron Source

Alberto D. Cencillo, Leonardo Concepción, Julián Luengo, Isaac Triguero (2026)

arXiv

Lightweight CNN-Based Anomaly Detection for High Voltage Converter Modulators in the Spallation Neutron Source

Cencillo, Alberto D., Concepción, Leonardo, Luengo, Julián, Triguero, Isaac (2026)

arXiv

An AI-ready, Polarized Electron-Positron Collision Dataset

Cheng, Chi Lung, Corrodi, Simon, Hobbs, T.J., Mete, Alaettin Serhan, Nachman, Benjamin (2026)

arXiv

Machine Learning for Process Optimization and Defect Detection in Metal Additive Manufacturing: A Critical Review of Algorithms, In-Situ Monitoring Strategies, and Quality Assurance Frameworks

Dara Jude, Ignatius Ekengwu, Anthonia Ilechukwu, Sunday Chimezie Anyaora, Augustine Uzochukwu Mmadumere (2026)

Journal of Mechanical Engineering and Manufacturing

Uncertainty Aware Functional Behavior Prediction and Material Fatigue Assessment for Circular Factory

Nehal Afifi, Mehdi Khabou, Victor Mas, Jonas Hemmerich, Patric Grauberger, Stefan Dietrich, Volker Schulze, Sven Matthiesen (2026)

arXiv

From satellite imagery to material outflows: master plan and deep learning-based quantification of demolition materials to support circular construction

Aiganym Kumisbek, Aslan Nauyryzbay, Ferhat Karaca, Mert Guney (2026)

Journal of Environmental Health Science and Engineering

Geiger Uses Machine Learning to Reduce Customer Churn in the Promotional Products Industry

Jan Schemm, Christian Schwarz, Marc Strickrodt (2026)
INFORMS Journal on Applied Analytics

Learning Dynamic Aperture from One-turn Maps

Derong Xu (2026)
arXiv

Integration of aerial photogrammetry and machine learning in the railway tracks infrastructure monitoring problem

Antoni Kopyt, Kacper Pawlowski (2026)
International Journal of Intelligent Unmanned Systems

Considerations for an Integrated Detector Design at FCC-ee: A Human-AI Exploration

Charles Young (2026)
arXiv

Training of ANN using Mountain Gazelle Optimizer for fire resistance of FRP-strengthened beams

Nima Khodadadi, M.Z. Naser, Francisco De Caso, Antonio Nanni, Khalid M Mosalam (2026)
Machine Learning in Civil Engineering and Infrastructure Development

Exploring the use of AI-generated counterfactual chest X-rays to enhance diagnostic learning in medical education

Greta Mohr, Yifei Zhu, Xujiong Ye, Marilyn Lennon, Calum MacLellan, John Maclay, David J. Lowe, Christopher Sainsbury, Feng Dong, David Lagnado (2026)
BMC Medical Education

A Correlational Study of Activity and Physics Learning Outcomes in the Implementation of Problem Based Learning among Grade XI Science Students at MAN 1 Mataram

Atika Wahyuni, Joni Rokhmat, I Wayan Gunada (2026)
International Journal of Contextual Science Education

Remaining useful life of power transformers using efficient surrogates and deep learning techniques

Lila Achour, Sebastian Rodriguez, Paul-Henri Langlois, Fikri Hafid, Chady Ghnatios, Xavier Kestelyn, Francisco Chinesta (2026)
Advanced Modeling and Simulation in Engineering Sciences

Machine-Learning-Based Prediction of Magnetic Recovery of Fine SUS304 Particles Using Magnetic Susceptibility Differences with a Minimum Effective Flux Density Design

Takashi Ohnishi, Keiichi Watanuki (2026)
Communications in Computer and Information Science

Attention-Driven Deep Reinforcement Learning for Energy-Efficient Trajectory and Beamforming Design in UAV-RIS Systems

Zhuo Fan, Lin Zhang, Tao Zhang, Yafeng Wang, Sheng Wu, Zhiyu Han (2026)
IEEE Internet of Things Journal

Violation detection in power operation sites based on multi-scale detection and few-shot learning

Yaokuan Wen, Jun Wang, Qiming Liu, Mo Zhou, Minzhe Tian (2026)
Frontiers in Artificial Intelligence

A Robust Supervised Learning Paradigm for Accurate Fault Detection and Spatio-Temporal Localization in Electrical Transmission Line

Md.Taukir Ahmed (2026)
Springer Science and Business Media LLC

Prioritizing Railway Level Crossing Safety Equipment Using Stabilized and Explainable Machine Learning: A Large-Scale Study of the French Railway Network

Nadège Reboul, Pierre-Antoine Laharotte, Rim Brahim, Philippe Bon, Simon Collart-Dutilleul, Nour-Eddin El Faouzi (2026)

Data Science for Transportation

Correction: Bhushan et al. Dynamic K-Decay Learning Rate Optimization for Deep Convolutional Neural Network to Estimate the State of Charge for Electric Vehicle Batteries. Energies 2024, 17, 3884

Neha Bhushan, Saad Mekhilef, Kok Soon Tey, Mohamed Shaaban, Mehdi Seyedmahmoudian, Alex Stojcevski (2026)
Energies

Dual-stage Multi-attention Deep Learning Framework for X-ray-based Classification of Rotator Cuff Tears

Chen-Chou Hsieh, Shang-Lin Hsieh, Yen-Yu Chen, Hsien-Chu Wu, Chwei-Shyong Tsai, Wei-Cheng Chang, Kuei-Wen Chen, Yi-Cheng Yang, Chin-Jung Hsu (2026)
Sensors and Materials

A hybrid monte Carlo-machine learning framework for neutron dose attenuation modeling in layered iron-concrete shields of a 1000 MeV proton accelerator

Demet Sarıyer, Ali Osman GÖKCAN (2026)
Annals of Nuclear Energy

Hardware-Software co-design of FETA: a Flexible Low Power AI accelerator for Time-series signals

Nazareno Sacchi, Stéphane Devise, Régis Cattenoz, Taekwang Jang (2026)
Proceedings of the 24th Annual International Conference on Mobile Systems, Applications and Services Workshops

STEM Learning in TTRPGs: What Is Being Assessed? How Hair Math Uses Storytelling and Play to Reimagine STEM Assessment

Sharon Lee De La Cruz, Sana Alimohamed (2026)
Proceedings of the 25th Annual ACM Interaction Design and Children Conference

Ein Framework zur Skalierung von Machine-Vision-Systemen in der Industrie

Jonas Werheid, Anja Koonen, Sylwia Olbrych, Anas Abdelrazeq, Robert H. Schmitt (2026)
Zeitschrift für wirtschaftlichen Fabrikbetrieb

SpiKint: Native Full-Integer Spiking Neural Networks Training with an Efficient CIM-based Accelerator

Peilin Chen, Xiaoxuan Yang (2026)
Proceedings of the Great Lakes Symposium on VLSI 2026

STING: A Stochastic In-DRAM Accelerator for Graph Neural Networks

Salma Afifi, Bipin Thapa Magar, Ishan Thakkar, Sudeep Pasricha (2026)
Proceedings of the Great Lakes Symposium on VLSI 2026

Unsupervised Learning for AC Optimal Power Flow with Fast Physics-Aware Layer

Jiebao Zhang, Haoyu Yan, Zhichao Sheng, Hongwen Yu, Shuang Ye, Haoyu Wang, Ye Shi (2026)
Proceedings of the 17th ACM International Conference on Future and Sustainable Energy Systems

Ultra-Peripheral Collisions as a Nuclear-Structure Interferometer with Interpretable Multitask Deep Learning

Jing-Zong Zhang, Wang-Mei Zha, Lingxiao Wang, Guo-Liang Ma (2026)
arXiv

Anticipating the Optimism Gap: Predicting Distribution-Shift Degradation of RF-Impairment Detectors from In-Distribution Statistics

Chakshu Baweja (2026)
arXiv

Learning missing physics from legacy simulators with alternating neural integrators

Hao Wang, Qinghe Wang, Caiyou Yuan, Kailiang Wu (2026)
Nature Communications

Reflective Learning-Based Particle Swarm Optimization for Feature Selection

Hongbo Zhang, Xiaofeng Yue, Xueliang Gao (2026)

International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems

Learning to Trigger: Reinforcement Learning at the Large Hadron Collider

Zixin Ding, Shaghayegh Emam, Giovanna Salvi, Cecilia Toscani, Abhijith Gandrakota, Jennifer Ngadiuba, Nhan Tran, Christian Herwig, David W. Miller, Yuxin Chen (2026)

arXiv

AI-Based Acoustic Rail Defect Detection System

K Bagyalaskhmi, C Harish, N A Harshitha, C Janani (2026)

Unknown Venue